var, x, y, z, w
$label$, L, $Next$, $Done$
lit
f
$dsort$, D
$tvar$, a, b
$index$, i, j, k, n

terminals	::=			
	|	$\rightarrow$		
	|	$\rightsquigarrow$		
	|	$\rightarrow$		
	|	$\mid$		
	|	$\vee$		
	|	$@$		
	|	$\multimap$		
	|	$\wedge$		
	|	$\Rightarrow$		
	|	$\in$		
	|	true		
	|	false		
blit	::=			boolean literals
	|	true		
	|	false		
eff, ϵ	::=			effects
	|	**0**		
	|	**1**		
	|	**S**		
kind	::=			kinds
	|	type		
	|	sort		
$\tau,\ A,\ B,\ C$	::=			sorts
	|	Int		integers
	|	Bool		booleans
	|	Ptr τ'		pointer
	|	$(\tau_1, .., \tau_n)$		tuple
	|	$D(\tau_1, .., \tau_n)$		datatype application
	|	Pred τ'		predicate sort
	|	a		type variable
	|	$[\gamma]\tau$	M	
$\overline{A}$	::=			type lists
	|	$[]$		
	|	$A{::}\overline{A}$		
F	::=			function signatures
	|	$\tau \rightarrow_{\mathit{eff}} \tau'$		
Γ	::=			contexts
	|	$\cdot$		
	|	$\Gamma, x{:}_{\mathit{eff}}\tau$		
	|	$\Gamma, a{:}\mathit{kind}$		
	|	Γ, Γ'		
	|	$\lvert\Delta\rvert$	M	
γ	::=			explicit substitutions
	|	$\cdot$		

	$\mid$	$\gamma, ce_1/x_1, \ldots, ce_n/x_n$		
	$\mid$	$\gamma, \tau_1/a_1, \ldots, \tau_n/a_n$		
	$\mid$	$\mathsf{id}(\Gamma)$	M	
	$\mid$	$\gamma; \gamma'$	M	
Σ_c	$::=$			core signatures
	$\mid$	$\cdot$		
	$\mid$	$\Sigma_c, f : F$		
	$\mid$	$\Sigma_c, \mathsf{fun}\ f(x : \tau) \rightarrow_{\mathit{eff}} \tau' = ce$		
	$\mid$	$\Sigma_c, \mathsf{sort}\ D(\overline{a}) = \{\overline{L : \tau}\}$		
	$\mid$	$\Sigma_c, \mathsf{type}\ D(\overline{a}) = \{\overline{L : A}\}$		
Σ_s	$::=$			surface signatures
	$\mid$	$\cdot$		
	$\mid$	$\Sigma_s, f : F$		
	$\mid$	$\Sigma_s, \mathsf{fun}\ f : \tau \rightarrow_{\mathit{eff}} \tau' = \{\overline{\mathbf{pat} \rightarrow \mathbf{se}}\}$		
	$\mid$	$\Sigma_s, \mathsf{sort}\ D(\overline{a}) = \{\overline{L : \tau}\}$		
	$\mid$	$\Sigma_s, \mathsf{type}\ D(\overline{a}) = \{\overline{L : A}\}$		
Σ_r	$::=$			refined signatures
	$\mid$	$\cdot$		
	$\mid$	$\Sigma_r, f : F$		
	$\mid$	$\Sigma_r, \mathsf{fun}\ f(x : \tau) \rightarrow_{\mathit{eff}} \tau' = ce$		
	$\mid$	$\Sigma_r, f : RF$		
	$\mid$	$\Sigma_r, \mathsf{sort}\ D(\overline{a}) = \{\overline{L : \tau}\}$		
	$\mid$	$\Sigma_r, \mathsf{type}\ D(\overline{a}) = \{\overline{L : A}\}$		
prim	$::=$			primitives
	$\mid$	**Add**		addition
	$\mid$	**Sub**		subtraction
	$\mid$	**Mul**		multiplication
	$\mid$	**Div**		division
	$\mid$	**And**		conjunction
	$\mid$	**Or**		disjunction
	$\mid$	**Not**		negation
	$\mid$	**Lt**		less than
	$\mid$	**Le**		less or equal
	$\mid$	**Gt**		greater than
	$\mid$	**Ge**		greater or equal
	$\mid$	**Eq** $[A]$		equality test
	$\mid$	**New** $[A]$		allocate
	$\mid$	**Del** $[A]$		deallocate
	$\mid$	**Get** $[A]$		dereference
	$\mid$	**Set** $[A]$		update
	$\mid$	**Own** $[A]$		ownership
ce	$::=$			core expressions
	$\mid$	x		
	$\mid$	*lit*		integer literal
	$\mid$	*blit*		boolean literal
	$\mid$	**let** $x = ce_1; ce_2$		
	$\mid$	$(ce_1, \ldots, ce_n)$		
	$\mid$	**let** $(x_1, \ldots, x_n) = ce_1; ce_2$		

	$\mid$	$L\ ce$	
	$\mid$	$\textbf{case}\ ce\ \textbf{of}\ \{L_1\ x_1 \to ce_1 \mid .. \mid L_n\ x_n \to ce_n\}$	
	$\mid$	$\textbf{iter}\ (x = ce_1)\{ce_2\}$	
	$\mid$	$prim\ ce$	
	$\mid$	$\textbf{if}\ ce_1\ \textbf{then}\ ce_2\ \textbf{else}\ ce_3$	conditional
	$\mid$	$ce : \tau$	
	$\mid$	$f\ ce$	
	$\mid$	$ce_1 \doteq ce_2$	
	$\mid$	$\textbf{take}\ x = ce_1; ce_2$	
	$\mid$	$\textbf{return}\ ce$	
	$\mid$	(ce)	
	$\mid$	$[\gamma]ce$	M
se	$::=$		surface expressions
	$\mid$	x	
	$\mid$	lit	integer literal
	$\mid$	$blit$	boolean literal
	$\mid$	$\textbf{let}\ pat = se_1; se_2$	
	$\mid$	$(se_1, .., se_n)$	
	$\mid$	$L\ se$	
	$\mid$	$\textbf{case}\ se\ \textbf{of}\ \{pat_1 \to se_1 \mid .. \mid pat_n \to se_n\}$	
	$\mid$	$\textbf{iter}\ (pat = se_1)\{se_2\}$	
	$\mid$	$prim\ se$	
	$\mid$	$\textbf{if}\ se_1\ \textbf{then}\ se_2\ \textbf{else}\ se_3$	conditional
	$\mid$	$se : \tau$	
	$\mid$	$f\ se$	
	$\mid$	$se_1 \doteq se_2$	
	$\mid$	$\textbf{take}\ pat = se_1; se_2$	
	$\mid$	$\textbf{return}\ se$	
$prog$	$::=$		surface programs
	$\mid$	$\mathsf{fun}\ f : \tau \to_{\mathit{eff}} \tau' = \{\overline{\textbf{pat} \to \textbf{se}}\}\quad prog$	
	$\mid$	$\textbf{sort}\ D(a_1, .., a_i) = \{L_1 : \tau_1 \mid .. \mid L_n : \tau_n\}prog$	
	$\mid$	$\textbf{type}\ D(a_1, .., a_i) = \{L_1 : A_1 \mid .. \mid L_n : A_n\}prog$	
	$\mid$	$\mathsf{main} : \tau_{\mathit{eff}} = se$	
$cprog$	$::=$		core programs
	$\mid$	$\mathsf{fun}\ f(x : \tau) \to_{\mathit{eff}} \tau' = ce\quad cprog$	
	$\mid$	$\textbf{sort}\ D(a_1, .., a_i) = \{L_1 : \tau_1 \mid .. \mid L_n : \tau_n\}cprog$	
	$\mid$	$\textbf{type}\ D(a_1, .., a_i) = \{L_1 : A_1 \mid .. \mid L_n : A_n\}cprog$	
	$\mid$	$\mathsf{main} : \tau_{\mathit{eff}} = ce$	
$rprog$	$::=$		refined programs
	$\mid$	$\textbf{sort}\ D(a_1, .., a_i) = \{L_1 : \tau_1 \mid .. \mid L_n : \tau_n\}rprog$	
	$\mid$	$\textbf{type}\ D(a_1, .., a_i) = \{L_1 : A_1 \mid .. \mid L_n : A_n\}rprog$	
	$\mid$	$\mathsf{fun}\ f(x : \tau) \to_{\mathit{eff}} \tau' = ce\quad rprog$	
	$\mid$	$\mathsf{fun}\ f(q) : Pf_1 \multimap Pf_2\ [\mathit{eff}] = crt\quad rprog$	
	$\mid$	$\mathsf{main} : Pf_{\mathit{eff}} = crt$	
pat	$::=$		patterns
	$\mid$	x	
	$\mid$	$L\ pat$	
	$\mid$	$(pat_1, .., pat_n)$	

β	$::=$		pattern bindings
	$\mid$	$\cdot$	
	$\mid$	$pat : \tau, \beta$	
	$\mid$	$pat_1 : \tau_1, \ldots, pat_n : \tau_n, \beta$	
	$\mid$	$pat : \tau$	
$\mathcal{E}$	$::=$		evaluation contexts
	$\mid$	$[\cdot]$	
	$\mid$	$\mathbf{let}\, x = ce; \mathcal{E}$	
$\overline{y}$	$::=$		scrutinee variables
	$\mid$	$\cdot$	
	$\mid$	$y, \overline{y}$	
	$\mid$	$y_1, \ldots, y_n, \overline{y}$	
M	$::=$		match branches
	$\mid$	$\cdot$	
	$\mid$	$\beta \to [\Gamma][\mathcal{E}]se \mid M$	
	$\mid$	$\beta \to [\Gamma][\mathcal{E}]se$	
$formula$	$::=$		
	$\mid$	$judgement$	
	$\mid$	$x{:}_{eff}\tau \in \Gamma$	
	$\mid$	$f{:}\tau \to_{eff} \tau' \in \Sigma_c$	
	$\mid$	A eqtype	
	$\mid$	$\mathbf{sort}\, D(a_1, \ldots, a_i)\, \mathbf{in}\, \Sigma_c$	
	$\mid$	$L{:}\tau \in \mathrm{ctors}(D) \in \Sigma_c$	
	$\mid$	$\mathbf{sort}\, D(a_1, \ldots, a_k) = \{L_1 : \tau_1 \mid \ldots \mid L_n : \tau_n\}\, \mathbf{in}\, \Sigma_c$	
	$\mid$	$\mathbf{type}\, D(a_1, \ldots, a_i)\, \mathbf{in}\, \Sigma_c$	
	$\mid$	$L{:}A \in \mathsf{tyctors}(D) \in \Sigma_c$	
	$\mid$	$\mathbf{type}\, D(a_1, \ldots, a_k) = \{L_1 : A_1 \mid \ldots \mid L_n : A_n\}\, \mathbf{in}\, \Sigma_c$	
	$\mid$	$\mathsf{type}\ D(a, b) = \{\mathsf{Next}{:}a \mid \mathsf{Done}{:}b\} \in \Sigma_c$	
	$\mid$	$a{:}kind \in \Gamma$	
	$\mid$	$kind \leq kind'$	
	$\mid$	$L \neq L'$	
	$\mid$	$M = \overline{pat{:}\tau, \cdot \to [\cdot][\cdot]\, se}$	
	$\mid$	$ce = \mathcal{E}[ce']$	
	$\mid$	$D \neq D'$	
	$\mid$	$x \neq y$	
	$\mid$	$a \neq b$	
	$\mid$	$n = k$	
	$\mid$	$eff = eff_1 \vee eff_2$	
	$\mid$	$eff = eff_1 \vee eff_2 \vee eff_3$	
	$\mid$	$eff = eff_1 \vee \ldots \vee eff_n$	
	$\mid$	$\Gamma = x_1 : \tau_1[eff_1], \ldots, x_n : \tau_n[eff_n]$	
	$\mid$	$\Gamma = \overline{a : k}$	
	$\mid$	$formula_1 \quad \ldots \quad formula_n$	
	$\mid$	$x{:}ce\ [\mathsf{log}] \in \Delta$	
	$\mid$	$x{:}ce@ce'\ [\mathsf{res}(u)] \in \Delta$	
	$\mid$	$x{:}_{eff}\tau \in \Delta$	
	$\mid$	$u \in \{1, ?\}$	
	$\mid$	$u \in \{0, ?\}$	
	$\mid$	$u = u_1 \sqcap u_2$	

	$\mid$	$\Delta' = \mathsf{affinize}(\Delta)$	
	$\mid$	$\mathsf{zero}(\Delta)$	
	$\mid$	$\Gamma = \lvert\Delta\rvert$	
	$\mid$	$\Gamma = \mathsf{Bind}(\cdot; Pf)$	
	$\mid$	$\mathcal{C} = \Delta \Rightarrow \mathcal{C}'$	
	$\mid$	$f(\mathbf{x} : \tau) \rightarrow_{\mathit{eff}} \tau' = ce \in \Sigma_c$	
	$\mid$	$\mathit{eff}' \in \{\mathbf{S}, \lfloor \mathit{eff} \rfloor\}$	
	$\mid$	$f{:}RF \in \Sigma_c$	
	$\mid$	$f{:}RF \in \Sigma_r$	
	$\mid$	$f{:}\tau \rightarrow_{\mathit{eff}} \tau' \in \Sigma_r$	
	$\mid$	$f(\mathbf{x} : \tau) \rightarrow_{\mathit{eff}} \tau' = ce \in \Sigma_r$	
	$\mid$	$\mathsf{len}(\Delta) = \mathsf{len}(\Delta')$	
	$\mid$	$A = \mathsf{Prod}(\overline{A})$	
	$\mid$	$\mathit{prim}{:}RF$	
u	$::=$		usage flags
	$\mid$	0	
	$\mid$	1	
	$\mid$	$?$	
Δ	$::=$		refined contexts
	$\mid$	$\cdot$	
	$\mid$	$\Delta, x{:}_{\mathit{eff}}\tau$	
	$\mid$	$\Delta, x{:}ce\ [\mathsf{log}]$	
	$\mid$	$\Delta, x{:}ce@ce'\ [\mathsf{res}(u)]$	
	$\mid$	Δ, Δ'	
$\overline{\Delta}$	$::=$		context lists
	$\mid$	$\Delta; \overline{\Delta}$	
	$\mid$	$\cdot$	
	$\mid$	$(\overline{\Delta})$	
	$\mid$	$\Delta_1; ..; \Delta_n$	
$\mathcal{C}$	$::=$		SMT constraints
	$\mid$	$\top$	
	$\mid$	$\mathcal{C}_1 \wedge \mathcal{C}_2$	
	$\mid$	$\forall x{:}\tau.\, \mathcal{C}$	
	$\mid$	$ce \Rightarrow \mathcal{C}$	
	$\mid$	ce	
	$\mid$	$\bot$	
	$\mid$	$(\mathcal{C})$	
Pf	$::=$		proof sorts
	$\mid$	$\cdot$	
	$\mid$	$x{:}_{\mathit{eff}}\tau, Pf$	
	$\mid$	$ce\ [\mathsf{log}], Pf$	
	$\mid$	$ce@ce'\ [\mathsf{res}], Pf$	
	$\mid$	$(y).ce\ [\mathsf{res}], Pf$	
	$\mid$	$[\gamma]Pf$	M
RF	$::=$		refined function types
	$\mid$	$Pf_1 \multimap Pf_2\ [\mathit{eff}]$	

q	$::=$		refined patterns
	$\mid$	$()$	
	$\mid$	(x, q)	
	$\mid$	$((x, y), q)$	
crt	$::=$		core refined terms
	$\mid$	$\mathbf{let}\ q = crt_1; crt_2$	
	$\mid$	$crt : Pf$	
	$\mid$	$prim\ rsp$	
	$\mid$	$f\ rsp$	
	$\mid$	(rsp)	
	$\mid$	$\mathbf{iter}\,[ce](q = crt_1)\{crt_2\}$	
	$\mid$	$\mathbf{if}\,[x]ce\,\mathbf{then}\,crt_1\,\mathbf{else}\,crt_2$	
	$\mid$	$\mathbf{case}\,[y]ce\,\mathbf{of}\,\{L_1\,x_1 \to crt_1 \mid .. \mid L_n\,x_n \to crt_n\}$	
	$\mid$	$\mathsf{exfalso}$	
	$\mid$	$\mathsf{open\text{-}take}\ rpf$	
rsp	$::=$		refined spines
	$\mid$	$\cdot$	
	$\mid$	ce, rsp	
	$\mid$	rpf, rsp	
	$\mid$	lpf, rsp	
lpf	$::=$		logical proof facts
	$\mid$	x	
	$\mid$	auto	
	$\mid$	$\mathsf{unfold}\ f(ce)$	
	$\mid$	$\mathsf{open\text{-}ret}\ rpf$	
	$\mid$	$lpf : ce$	
rpf	$::=$		resource proof facts
	$\mid$	x	
	$\mid$	$\mathsf{make\text{-}ret}\ lpf$	
	$\mid$	$\mathsf{make\text{-}take}\ crt$	
	$\mid$	$rpf : ce@ce'$	
$Jtype$	$::=$		
	$\mid$	$kind \leq kind'$	
	$\mid$	$\Sigma_c; \Gamma \vdash \tau{:}kind$	
	$\mid$	$\Sigma_c; \Gamma; D'(\overline{a}) \vdash, \mathsf{guarded}$	
	$\mid$	$\Sigma_c \vdash \mathsf{sort}\ D(\overline{a}) = \{\overline{L : \tau}\}\ \mathsf{ok}$	
	$\mid$	$\Sigma_c \vdash \mathsf{type}\ D(\overline{a}) = \{\overline{L : A}\}\ \mathsf{ok}$	
	$\mid$	$\Sigma_c; \Gamma \vdash \gamma : \Gamma'$	
	$\mid$	$\gamma = \mathsf{id}(\Gamma)$	
	$\mid$	$\tau' = [\gamma]\tau$	
	$\mid$	$ce' = [\gamma]ce$	
	$\mid$	$Pf' = [\gamma]Pf$	
	$\mid$	$\gamma'' = \gamma\,;\gamma'$	
	$\mid$	$A\ \mathsf{eqtype}$	
	$\mid$	$eff \leq eff'$	
	$\mid$	$eff' = \lfloor eff \rfloor$	
	$\mid$	$eff = eff_1 \vee eff_2$	
	$\mid$	$prim{:}\tau \to_{eff} \tau'$	

	$\mid$	$\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} ce \Rightarrow \tau$
	$\mid$	$\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} ce \Leftarrow \tau$
	$\mid$	$\Sigma_c \vdash cprog$
$Jcore$	$::=$	
	$\mid$	$L : \tau \,\mathbf{in}\, D(\tau_1, .., \tau_i) \,\mathbf{in}\, \Sigma_c$
	$\mid$	$D(a_1, .., a_k) = \{L_1 : \tau_1 \mid .. \mid L_n : \tau_n\} \in \Sigma_c$
$Jelaborate$	$::=$	
	$\mid$	$\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} se \Rightarrow \tau \rightsquigarrow ce$
	$\mid$	$\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} se \Leftarrow \tau \rightsquigarrow ce$
	$\mid$	$\Sigma_s \vdash prog$ $\rightsquigarrow \Sigma_c \vdash cprog$
$Jmatch$	$::=$	
	$\mid$	M has_con
	$\mid$	M has_tup
	$\mid$	$M/var \overset{y}{\rightsquigarrow}_{\mathit{eff}} M'$
	$\mid$	$M/L{:}\tau \overset{y}{\rightsquigarrow}_{\mathit{eff}} M'$
	$\mid$	$M/\mathbf{tup} \overset{y}{\rightsquigarrow}_{\mathit{eff}} M'$
	$\mid$	$\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \overline{y}; M \Leftarrow_\epsilon \tau \rightsquigarrow ce$
	$\mid$	$\mathcal{E}[ce] \rightsquigarrow ce'$
$Jrefine$	$::=$	
	$\mid$	$\Sigma_r \vdash \Delta$ wf
	$\mid$	$\Gamma = \lvert\Delta\rvert$
	$\mid$	$\Delta' = \mathsf{affinize}(\Delta)$
	$\mid$	$\mathsf{zero}(\Delta)$
	$\mid$	$\mathcal{C} = \Delta \Rightarrow \mathcal{C}'$
	$\mid$	$\Sigma_r; \Gamma \vdash \mathcal{C}$ wf
	$\mid$	$\Sigma_r; \Gamma \vdash_{\mathit{eff}} Pf$ wf
	$\mid$	$\Sigma_r \vdash RF$ wf
	$\mid$	$\Sigma_r \vdash f{:}RF$
	$\mid$	$\overline{A} = \{Pf\}$
	$\mid$	$\Sigma_c = \mathsf{Comp}(\Sigma_r)$
	$\mid$	$\vdash \Sigma_r$ wf
	$\mid$	$\Sigma_r; \Delta \vdash lpf \Rightarrow ce \dashv \Delta' \rightsquigarrow \mathcal{C}$
	$\mid$	$\Sigma_r; \Delta \vdash lpf \Leftarrow ce \dashv \Delta' \rightsquigarrow \mathcal{C}$
	$\mid$	$\Sigma_r; \Delta \vdash x : ce@ce' \dashv \Delta'$
	$\mid$	$\Sigma_r; \Delta \vdash rpf \Rightarrow ce@ce' \dashv \Delta' \rightsquigarrow \mathcal{C}$
	$\mid$	$\Sigma_r; \Delta \vdash rpf \Leftarrow ce@ce' \dashv \Delta' \rightsquigarrow \mathcal{C}$
	$\mid$	$\Sigma_r; \Delta \vdash Pf_1 = Pf_2 \rightsquigarrow \mathcal{C}$
	$\mid$	$\Sigma_r \vdash \Delta_0 \sqcap \Delta_1 = \Delta$
	$\mid$	$\Sigma_r; \Gamma \vdash q : Pf \dashv \Delta$
	$\mid$	$\Sigma_r; \Delta \vdash_{\mathit{eff}} crt \Rightarrow Pf \dashv \Delta' \rightsquigarrow \mathcal{C}$
	$\mid$	$\Sigma_r; \Delta \vdash_{\mathit{eff}} y{:}ce$ is $L\,x \rightarrow crt \Leftarrow Pf \dashv \Delta' \rightsquigarrow \mathcal{C}$
	$\mid$	$\Sigma_r; \Delta \vdash_{\mathit{eff}} crt \Leftarrow Pf \dashv \Delta' \rightsquigarrow \mathcal{C}$
	$\mid$	$\Sigma_r; \Delta \vdash_{\mathit{eff}} rsp : Pf_1 \multimap Pf_2 \gg Pf \dashv \Delta' \rightsquigarrow \mathcal{C}$
	$\mid$	$\Sigma_r; \Delta \vdash_{\mathit{eff}} rsp : Pf \dashv \Delta' \rightsquigarrow \mathcal{C}$
	$\mid$	$\Sigma_r \vdash rprog \dashv \Sigma'_r \rightsquigarrow \mathcal{C}$
	$\mid$	$\Sigma_r \vdash \sqcap\overline{\Delta} = \Delta$
$judgement$	$::=$	
	$\mid$	$Jtype$

	|	*Jcore*
	|	*Jelaborate*
	|	*Jmatch*
	|	*Jrefine*
user_syntax	::=	
	|	*var*
	|	*label*
	|	*lit*
	|	f
	|	*dsort*
	|	*tvar*
	|	*index*
	|	*terminals*
	|	*blit*
	|	*eff*
	|	*kind*
	|	τ
	|	$\overline{A}$
	|	F
	|	Γ
	|	γ
	|	Σ_c
	|	Σ_s
	|	Σ_r
	|	*prim*
	|	*ce*
	|	*se*
	|	*prog*
	|	*cprog*
	|	*rprog*
	|	*pat*
	|	β
	|	$\mathcal{E}$
	|	$\overline{y}$
	|	M
	|	*formula*
	|	u
	|	Δ
	|	$\overline{\Delta}$
	|	$\mathcal{C}$
	|	*Pf*
	|	*RF*
	|	q
	|	*crt*
	|	*rsp*
	|	*lpf*
	|	*rpf*

$\boxed{kind \leq kind'}$

$$\frac{}{\mathsf{type} \leq \mathsf{type}} \quad \text{SK_TYPE_TYPE}$$

$$\frac{}{\mathsf{type} \leq \mathsf{sort}} \quad \text{SK_TYPE_SORT}$$

$$\frac{}{\mathsf{sort} \leq \mathsf{sort}} \quad \text{SK_SORT_SORT}$$

$\boxed{\Sigma_c; \Gamma \vdash \tau{:}\mathit{kind}}$

$$\frac{}{\Sigma_c; \Gamma \vdash \mathsf{Int}{:}\mathit{kind}} \quad \text{KWF_INT}$$

$$\frac{}{\Sigma_c; \Gamma \vdash \mathsf{Bool}{:}\mathit{kind}} \quad \text{KWF_BOOL}$$

$$\frac{\Sigma_c; \Gamma \vdash \tau{:}\mathsf{type}}{\Sigma_c; \Gamma \vdash \mathsf{Ptr}\ \tau{:}\mathit{kind}} \quad \text{KWF_PTR}$$

$$\frac{\Sigma_c; \Gamma \vdash \tau_1{:}\mathit{kind} \quad .. \quad \Sigma_c; \Gamma \vdash \tau_n{:}\mathit{kind}}{\Sigma_c; \Gamma \vdash (\tau_1, .., \tau_n){:}\mathit{kind}} \quad \text{KWF_RECORD}$$

$$\frac{\begin{array}{c}\textbf{type}\, D(a_1, .., a_k)\, \textbf{in}\, \Sigma_c \quad n = k \\ \Sigma_c; \Gamma \vdash \tau_1{:}\mathsf{type} \quad .. \quad \Sigma_c; \Gamma \vdash \tau_n{:}\mathsf{type}\end{array}}{\Sigma_c; \Gamma \vdash D(\tau_1, .., \tau_n){:}\mathsf{type}} \quad \text{KWF_TYPE_APP}$$

$$\frac{\begin{array}{c}\textbf{sort}\, D(a_1, .., a_k)\, \textbf{in}\, \Sigma_c \quad n = k \\ \Sigma_c; \Gamma \vdash \tau_1{:}\mathsf{sort} \quad .. \quad \Sigma_c; \Gamma \vdash \tau_n{:}\mathsf{sort}\end{array}}{\Sigma_c; \Gamma \vdash D(\tau_1, .., \tau_n){:}\mathsf{sort}} \quad \text{KWF_SORT_APP}$$

$$\frac{\Sigma_c; \Gamma \vdash \tau{:}\mathsf{sort}}{\Sigma_c; \Gamma \vdash \mathsf{Pred}\ \tau{:}\mathsf{sort}} \quad \text{KWF_PRED}$$

$$\frac{a{:}\mathit{kind}' \in \Gamma \quad \mathit{kind}' \leq \mathit{kind}}{\Sigma_c; \Gamma \vdash a{:}\mathit{kind}} \quad \text{KWF_TVAR}$$

$\boxed{\Sigma_c; \Gamma; D'(\overline{a}) \vdash,\ \mathsf{guarded}}$

$$\frac{}{\Sigma_c; \Gamma; D'(\overline{a}) \vdash,\ \mathsf{guarded}} \quad \text{TWFG_INT}$$

$$\frac{}{\Sigma_c; \Gamma; D'(\overline{a}) \vdash,\ \mathsf{guarded}} \quad \text{TWFG_BOOL}$$

$$\frac{\Sigma_c, \mathsf{type}\ D'(\overline{a}) = \{\overline{L : A}\}; \Gamma \vdash \tau{:}\mathsf{type}}{\Sigma_c; \Gamma; D'(\overline{a}) \vdash,\ \mathsf{guarded}} \quad \text{TWFG_PTR}$$

$$\frac{\Sigma_c; \Gamma; D'(\overline{a}) \vdash,\ \mathsf{guarded} \quad .. \quad \Sigma_c; \Gamma; D'(\overline{a}) \vdash,\ \mathsf{guarded}}{\Sigma_c; \Gamma; D'(\overline{a}) \vdash,\ \mathsf{guarded}} \quad \text{TWFG_RECORD}$$

$$\frac{\begin{array}{c}\textbf{type}\, D(b_1, .., b_k)\, \textbf{in}\, \Sigma_c \quad n = k \quad D \neq D' \\ \Sigma_c; \Gamma \vdash \tau_1{:}\mathsf{type} \quad .. \quad \Sigma_c; \Gamma \vdash \tau_n{:}\mathsf{type}\end{array}}{\Sigma_c; \Gamma; D'(\overline{a}) \vdash,\ \mathsf{guarded}} \quad \text{TWFG_APP}$$

$$\frac{a{:}\mathit{kind} \in \Gamma}{\Sigma_c; \Gamma; D'(\overline{a}) \vdash,\ \mathsf{guarded}} \quad \text{TWFG_TVAR}$$

$\boxed{\Sigma_c \vdash \mathsf{sort}\ D(\overline{a}) = \{\overline{L : \tau}\}\ \mathsf{ok}}$

$$\frac{\begin{array}{c}\Gamma = \overline{a : k} \\ \Sigma_c, \mathsf{sort}\ D(\overline{a}) = \{\overline{L : \tau}\}; \Gamma \vdash \tau_1{:}\mathsf{sort} \quad .. \quad \Sigma_c, \mathsf{sort}\ D(\overline{a}) = \{\overline{L : \tau}\}; \Gamma \vdash \tau_n{:}\mathsf{sort}\end{array}}{\Sigma_c \vdash \mathsf{sort}\ D(\overline{a}) = \{\overline{L : \tau}\}\ \mathsf{ok}} \quad \text{SDWF_OK}$$

$\boxed{\Sigma_c \vdash \mathsf{type}\ D(\overline{a}) = \{\overline{L : A}\}\ \mathsf{ok}}$

$$\frac{\begin{array}{c}\Gamma = \overline{a : k} \\ \Sigma_c; \Gamma; D(\overline{a}) \vdash,\ \mathsf{guarded} \quad .. \quad \Sigma_c; \Gamma; D(\overline{a}) \vdash,\ \mathsf{guarded}\end{array}}{\Sigma_c \vdash \mathsf{type}\ D(\overline{a}) = \{\overline{L : A}\}\ \mathsf{ok}} \quad \text{TDWF_OK}$$

$\boxed{\Sigma_c; \Gamma \vdash \gamma : \Gamma'}$

$$\frac{}{\Sigma_c; \Gamma \vdash \cdot : \cdot} \quad \textsc{SubWf_empty}$$

$$\frac{\Sigma_c; \Gamma \vdash \gamma : \Gamma' \quad \Sigma_c; \Gamma \vdash_{\mathit{eff}} ce \Rightarrow \tau}{\Sigma_c; \Gamma \vdash \gamma, ce/x : \Gamma', x{:}_{\mathit{eff}}\tau} \quad \textsc{SubWf_term}$$

$$\frac{\Sigma_c; \Gamma \vdash \gamma : \Gamma' \quad \Sigma_c; \Gamma \vdash \tau{:}\mathit{kind}}{\Sigma_c; \Gamma \vdash \gamma, \tau/a : \Gamma', a{:}\mathit{kind}} \quad \textsc{SubWf_type}$$

$\boxed{\gamma = \mathsf{id}(\Gamma)}$

$$\frac{}{\cdot = \mathsf{id}(\cdot)} \quad \textsc{IdDef_empty}$$

$$\frac{\gamma = \mathsf{id}(\Gamma)}{\gamma, x/x = \mathsf{id}(\Gamma, x{:}_{\mathit{eff}}\tau)} \quad \textsc{IdDef_term}$$

$$\frac{\gamma = \mathsf{id}(\Gamma)}{\gamma, a/a = \mathsf{id}(\Gamma, a{:}\mathit{kind})} \quad \textsc{IdDef_type}$$

$\boxed{\tau' = [\gamma]\tau}$

$$\frac{}{\mathsf{Int} = [\gamma]\mathsf{Int}} \quad \textsc{SS_int}$$

$$\frac{}{\mathsf{Bool} = [\gamma]\mathsf{Bool}} \quad \textsc{SS_bool}$$

$$\frac{\tau' = [\gamma]\tau}{\mathsf{Ptr}\ \tau' = [\gamma]\mathsf{Ptr}\ \tau} \quad \textsc{SS_ptr}$$

$$\frac{\tau'_1 = [\gamma]\tau_1 \quad .. \quad \tau'_n = [\gamma]\tau_n}{(\tau'_1, .., \tau'_n) = [\gamma](\tau_1, .., \tau_n)} \quad \textsc{SS_record}$$

$$\frac{\tau'_1 = [\gamma]\tau_1 \quad .. \quad \tau'_n = [\gamma]\tau_n}{D(\tau'_1, .., \tau'_n) = [\gamma]D(\tau_1, .., \tau_n)} \quad \textsc{SS_app}$$

$$\frac{\tau' = [\gamma]\tau}{\mathsf{Pred}\ \tau' = [\gamma]\mathsf{Pred}\ \tau} \quad \textsc{SS_pred}$$

$$\frac{}{\tau = [\gamma, \tau/a]a} \quad \textsc{SS_tvar_here}$$

$$\frac{a \neq b \quad \tau' = [\gamma]b}{\tau' = [\gamma, \tau/a]b} \quad \textsc{SS_tvar_skip_ty}$$

$$\frac{\tau' = [\gamma]b}{\tau' = [\gamma, ce/x]b} \quad \textsc{SS_tvar_skip_tm}$$

$\boxed{ce' = [\gamma]ce}$

$$\frac{}{ce = [\gamma, ce/x]x} \quad \textsc{TS_var_here}$$

$$\frac{x \neq y \quad ce' = [\gamma]x}{ce' = [\gamma, ce/y]x} \quad \textsc{TS_var_skip_tm}$$

$$\frac{ce' = [\gamma]x}{ce' = [\gamma, \tau/a]x} \quad \textsc{TS_var_skip_ty}$$

$$\frac{}{lit = [\gamma]lit} \quad \textsc{TS_intlit}$$

$$\frac{}{blit = [\gamma]blit} \quad \textsc{TS_boollit}$$

$$\frac{ce_1' = [\gamma]ce_1 \quad ce_2' = [\gamma, x/x]ce_2}{\textbf{let}\, x = ce_1'; ce_2' = [\gamma]\textbf{let}\, x = ce_1; ce_2} \quad \text{TS_LET}$$

$$\frac{ce_1' = [\gamma]ce_1 \quad .. \quad ce_n' = [\gamma]ce_n}{(ce_1', .., ce_n') = [\gamma](ce_1, .., ce_n)} \quad \text{TS_TUPLE}$$

$$\frac{ce_1' = [\gamma]ce_1 \quad ce_2' = [\gamma, x_1/x_1, .., x_n/x_n]ce_2}{\textbf{let}\,(x_1, .., x_n) = ce_1'; ce_2' = [\gamma]\textbf{let}\,(x_1, .., x_n) = ce_1; ce_2} \quad \text{TS_LETTUPLE}$$

$$\frac{ce' = [\gamma]ce}{L\, ce' = [\gamma]L\, ce} \quad \text{TS_INJECT}$$

$$\frac{ce' = [\gamma]ce \quad ce_1' = [\gamma, x_1/x_1]ce_1 \quad .. \quad ce_n' = [\gamma, x_n/x_n]ce_n}{\textbf{case}\, ce'\, \textbf{of}\, \{L_1\, x_1 \to ce_1' | .. | L_n\, x_n \to ce_n'\} = [\gamma]\textbf{case}\, ce\, \textbf{of}\, \{L_1\, x_1 \to ce_1 | .. | L_n\, x_n \to ce_n\}} \quad \text{TS_CASE}$$

$$\frac{ce_1' = [\gamma]ce_1 \quad ce_2' = [\gamma, x/x]ce_2}{\textbf{iter}\,(x = ce_1')\{ce_2'\} = [\gamma]\textbf{iter}\,(x = ce_1)\{ce_2\}} \quad \text{TS_ITER}$$

$$\frac{ce' = [\gamma]ce}{prim\, ce' = [\gamma]prim\, ce} \quad \text{TS_PRIM}$$

$$\frac{ce_1' = [\gamma]ce_1 \quad ce_2' = [\gamma]ce_2 \quad ce_3' = [\gamma]ce_3}{\textbf{if}\, ce_1'\, \textbf{then}\, ce_2'\, \textbf{else}\, ce_3' = [\gamma]\textbf{if}\, ce_1\, \textbf{then}\, ce_2\, \textbf{else}\, ce_3} \quad \text{TS_IF}$$

$$\frac{ce' = [\gamma]ce \quad \tau' = [\gamma]\tau}{ce' : \tau' = [\gamma]ce : \tau} \quad \text{TS_ANNOT}$$

$$\frac{ce' = [\gamma]ce}{f\, ce' = [\gamma]f\, ce} \quad \text{TS_CALL}$$

$$\frac{ce_1' = [\gamma]ce_1 \quad ce_2' = [\gamma]ce_2}{ce_1' \doteq ce_2' = [\gamma]ce_1 \doteq ce_2} \quad \text{TS_EQ}$$

$$\frac{ce_1' = [\gamma]ce_1 \quad ce_2' = [\gamma, x/x]ce_2}{\textbf{take}\, x = ce_1'; ce_2' = [\gamma]\textbf{take}\, x = ce_1; ce_2} \quad \text{TS_TAKE}$$

$$\frac{ce' = [\gamma]ce}{\textbf{return}\, ce' = [\gamma]\textbf{return}\, ce} \quad \text{TS_RETURN}$$

$\boxed{Pf' = [\gamma]Pf}$

$$\frac{}{\cdot = [\gamma]\cdot} \quad \text{PS_NIL}$$

$$\frac{\tau' = [\gamma]\tau \quad Pf' = [\gamma, x/x]Pf}{x{:}_{\mathit{eff}}\,\tau', Pf' = [\gamma]x{:}_{\mathit{eff}}\,\tau, Pf} \quad \text{PS_COMP}$$

$$\frac{ce' = [\gamma]ce \quad Pf' = [\gamma]Pf}{ce'\ [\mathsf{log}], Pf' = [\gamma]ce\ [\mathsf{log}], Pf} \quad \text{PS_LOG}$$

$$\frac{ce_1' = [\gamma]ce_1 \quad ce_2' = [\gamma]ce_2 \quad Pf' = [\gamma]Pf}{ce_1'@ce_2'\ [\mathsf{res}], Pf' = [\gamma]ce_1@ce_2\ [\mathsf{res}], Pf} \quad \text{PS_RES}$$

$$\frac{ce' = [\gamma, y/y]ce \quad Pf' = [\gamma]Pf}{(y).ce'\ [\mathsf{res}], Pf' = [\gamma](y).ce\ [\mathsf{res}], Pf} \quad \text{PS_DEPRES}$$

$\boxed{\gamma'' = \gamma\,;\gamma'}$

$$\frac{}{\cdot = \gamma\,;\cdot} \quad \text{CompDef_empty}$$

$$\frac{\gamma'' = \gamma\,;\gamma' \quad ce' = [\gamma]ce}{\gamma'', ce'/x = \gamma\,;\gamma', ce/x} \quad \text{CompDef_term}$$

$$\frac{\gamma'' = \gamma\,;\gamma' \quad \tau' = [\gamma]\tau}{\gamma'', \tau'/a = \gamma\,;\gamma', \tau/a} \quad \textsc{CompDef_type}$$

$\boxed{A \text{ eqtype}}$

$$\frac{}{\mathsf{Int\ eqtype}} \quad \textsc{EqTy_int}$$

$$\frac{}{\mathsf{Bool\ eqtype}} \quad \textsc{EqTy_bool}$$

$$\frac{}{\mathsf{Ptr}\ A\ \mathsf{eqtype}} \quad \textsc{EqTy_ptr}$$

$\boxed{\mathit{eff} \leq \mathit{eff}'}$

$$\frac{}{\mathbf{0} \leq \mathbf{0}} \quad \textsc{Sub_pure_pure}$$

$$\frac{}{\mathbf{0} \leq \mathbf{1}} \quad \textsc{Sub_pure_impure}$$

$$\frac{}{\mathbf{1} \leq \mathbf{1}} \quad \textsc{Sub_impure_impure}$$

$$\frac{}{\mathbf{0} \leq \mathbf{S}} \quad \textsc{Sub_pure_spec}$$

$$\frac{}{\mathbf{S} \leq \mathbf{S}} \quad \textsc{Sub_spec_spec}$$

$\boxed{\mathit{eff}' = \lfloor \mathit{eff} \rfloor}$

$$\frac{}{\mathbf{0} = \lfloor \mathbf{0} \rfloor} \quad \textsc{Pur_pure}$$

$$\frac{}{\mathbf{0} = \lfloor \mathbf{1} \rfloor} \quad \textsc{Pur_impure}$$

$$\frac{}{\mathbf{S} = \lfloor \mathbf{S} \rfloor} \quad \textsc{Pur_spec}$$

$\boxed{\mathit{eff} = \mathit{eff}_1 \vee \mathit{eff}_2}$

$$\frac{}{\mathit{eff} = \mathit{eff} \vee \mathit{eff}} \quad \textsc{Join_refl}$$

$$\frac{}{\mathbf{1} = \mathbf{0} \vee \mathbf{1}} \quad \textsc{Join_pure_impure}$$

$$\frac{}{\mathbf{1} = \mathbf{1} \vee \mathbf{0}} \quad \textsc{Join_impure_pure}$$

$$\frac{}{\mathbf{S} = \mathbf{0} \vee \mathbf{S}} \quad \textsc{Join_pure_spec}$$

$$\frac{}{\mathbf{S} = \mathbf{S} \vee \mathbf{0}} \quad \textsc{Join_spec_pure}$$

$\boxed{\mathit{prim}{:}\tau \rightarrow_{\mathit{eff}} \tau'}$

$$\frac{}{\mathbf{Add}{:}(\mathsf{Int}, \mathsf{Int}) \rightarrow_{\mathbf{0}} \mathsf{Int}} \quad \textsc{PT_add}$$

$$\frac{}{\mathbf{Sub}{:}(\mathsf{Int}, \mathsf{Int}) \rightarrow_{\mathbf{0}} \mathsf{Int}} \quad \textsc{PT_sub}$$

$$\frac{}{\mathbf{Mul}{:}(\mathsf{Int}, \mathsf{Int}) \rightarrow_{\mathbf{0}} \mathsf{Int}} \quad \textsc{PT_mul}$$

$$\frac{}{\mathbf{Div}{:}(\mathsf{Int}, \mathsf{Int}) \rightarrow_{\mathbf{1}} \mathsf{Int}} \quad \textsc{PT_div}$$

$$\frac{}{\mathbf{And}{:}(\mathsf{Bool}, \mathsf{Bool}) \rightarrow_{\mathbf{0}} \mathsf{Bool}} \quad \textsc{PT_and}$$

$$\frac{}{\mathbf{Or}{:}(\mathsf{Bool}, \mathsf{Bool}) \rightarrow_{\mathbf{0}} \mathsf{Bool}} \quad \text{PT_OR}$$

$$\frac{}{\mathbf{Not}{:}\mathsf{Bool} \rightarrow_{\mathbf{0}} \mathsf{Bool}} \quad \text{PT_NOT}$$

$$\frac{}{\mathbf{Lt}{:}(\mathsf{Int}, \mathsf{Int}) \rightarrow_{\mathbf{0}} \mathsf{Bool}} \quad \text{PT_LT}$$

$$\frac{}{\mathbf{Le}{:}(\mathsf{Int}, \mathsf{Int}) \rightarrow_{\mathbf{0}} \mathsf{Bool}} \quad \text{PT_LE}$$

$$\frac{}{\mathbf{Gt}{:}(\mathsf{Int}, \mathsf{Int}) \rightarrow_{\mathbf{0}} \mathsf{Bool}} \quad \text{PT_GT}$$

$$\frac{}{\mathbf{Ge}{:}(\mathsf{Int}, \mathsf{Int}) \rightarrow_{\mathbf{0}} \mathsf{Bool}} \quad \text{PT_GE}$$

$$\frac{A \ \mathsf{eqtype}}{\mathbf{Eq}\,[A]{:}(A, A) \rightarrow_{\mathbf{0}} \mathsf{Bool}} \quad \text{PT_EQ}$$

$$\frac{}{\mathbf{New}\,[A]{:}A \rightarrow_{\mathbf{1}} \mathsf{Ptr}\ A} \quad \text{PT_NEW}$$

$$\frac{}{\mathbf{Del}\,[A]{:}\mathsf{Ptr}\ A \rightarrow_{\mathbf{1}} (\,)} \quad \text{PT_DEL}$$

$$\frac{}{\mathbf{Get}\,[A]{:}\mathsf{Ptr}\ A \rightarrow_{\mathbf{1}} A} \quad \text{PT_GET}$$

$$\frac{}{\mathbf{Set}\,[A]{:}(\mathsf{Ptr}\ A, A) \rightarrow_{\mathbf{1}} (\,)} \quad \text{PT_SET}$$

$$\frac{}{\mathbf{Own}\,[A]{:}\mathsf{Ptr}\ A \rightarrow_{\mathbf{S}} \mathsf{Pred}\ (A)} \quad \text{PT_OWN}$$

$\boxed{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathit{ce} \Rightarrow \tau}$

$$\frac{}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathit{lit} \Rightarrow \mathsf{Int}} \quad \text{SYN_INT}$$

$$\frac{}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathit{blit} \Rightarrow \mathsf{Bool}} \quad \text{SYN_BOOL}$$

$$\frac{x{:}_{\mathit{eff}'}\tau \in \Gamma \quad \mathit{eff}' \leq \mathit{eff}_0}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} x \Rightarrow \tau} \quad \text{SYN_VAR}$$

$$\frac{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathit{ce} \Leftarrow \tau}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathit{ce} : \tau \Rightarrow \tau} \quad \text{SYN_ANNOT}$$

$$\frac{\begin{array}{l} \mathit{prim}{:}\tau \rightarrow_{\mathit{eff}'} \tau' \\ \mathit{eff}' \leq \mathit{eff}_0 \\ \mathit{eff}'' = \lfloor \mathit{eff}_0 \rfloor \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}''} \mathit{ce} \Leftarrow \tau \end{array}}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathit{prim}\ \mathit{ce} \Rightarrow \tau'} \quad \text{SYN_APP}$$

$$\frac{\begin{array}{l} f{:}\tau \rightarrow_{\mathit{eff}'} \tau' \in \Sigma_c \\ \mathit{eff}' \leq \mathit{eff}_0 \\ \mathit{eff}'' = \lfloor \mathit{eff}_0 \rfloor \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}''} \mathit{ce} \Leftarrow \tau \end{array}}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} f\ \mathit{ce} \Rightarrow \tau'} \quad \text{SYN_CALL}$$

$$\frac{\begin{array}{l} A \ \mathsf{eqtype} \\ \mathit{eff}' = \lfloor \mathit{eff}_0 \rfloor \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}'} \mathit{ce}_1 \Rightarrow A \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}'} \mathit{ce}_2 \Leftarrow A \end{array}}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathit{ce}_1 \doteq \mathit{ce}_2 \Rightarrow \mathsf{Bool}} \quad \text{SYN_EQ}$$

$\boxed{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathit{ce} \Leftarrow \tau}$

$$\frac{\begin{array}{l}\Sigma_c;\Gamma \vdash_{\mathit{eff}_0} ce_1 \Rightarrow \tau \\ \epsilon = \lfloor \mathit{eff}_0 \rfloor \\ \Sigma_c;\Gamma, x{:}_\epsilon \tau \vdash_{\mathit{eff}_0} ce_2 \Leftarrow \tau'\end{array}}{\Sigma_c;\Gamma \vdash_{\mathit{eff}_0} \mathbf{let}\, x = ce_1; ce_2 \Leftarrow \tau'} \quad \textsc{Chk_let}$$

$$\frac{\Sigma_c;\Gamma \vdash_{\mathit{eff}_0} ce_1 \Leftarrow \tau_1 \quad .. \quad \Sigma_c;\Gamma \vdash_{\mathit{eff}_0} ce_n \Leftarrow \tau_n}{\Sigma_c;\Gamma \vdash_{\mathit{eff}_0} (ce_1, .., ce_n) \Leftarrow (\tau_1, .., \tau_n)} \quad \textsc{Chk_tuple}$$

$$\frac{\begin{array}{l}\epsilon = \lfloor \mathit{eff}_0 \rfloor \\ \Sigma_c;\Gamma \vdash_\epsilon ce_1 \Rightarrow (\tau_1, .., \tau_n) \\ \Gamma' = x_1 : \tau_1[\epsilon], .., x_n : \tau_n[\epsilon] \\ \Sigma_c;\Gamma, \Gamma' \vdash_{\mathit{eff}_0} ce_2 \Leftarrow \tau'\end{array}}{\Sigma_c;\Gamma \vdash_{\mathit{eff}_0} \mathbf{let}\, (x_1, .., x_n) = ce_1; ce_2 \Leftarrow \tau'} \quad \textsc{Chk_lettuple}$$

$$\frac{\begin{array}{l}L : \tau\, \mathbf{in}\, D(\tau_1, .., \tau_i)\, \mathbf{in}\, \Sigma_c \\ \mathit{eff}' = \lfloor \mathit{eff}_0 \rfloor \\ \Sigma_c;\Gamma \vdash_{\mathit{eff}'} ce \Leftarrow \tau\end{array}}{\Sigma_c;\Gamma \vdash_{\mathit{eff}_0} L\, ce \Leftarrow D(\tau_1, .., \tau_i)} \quad \textsc{Chk_inject}$$

$$\frac{\begin{array}{l}\mathit{eff}' = \lfloor \mathit{eff}_0 \rfloor \\ \Sigma_c;\Gamma \vdash_{\mathit{eff}'} ce \Rightarrow D(\tau_1, .., \tau_i) \\ D(a_1, .., a_k) = \{L_1 : \tau_1' | .. | L_n : \tau_n'\} \in \Sigma_c \\ L_1 : \tau_1''\, \mathbf{in}\, D(\tau_1, .., \tau_i)\, \mathbf{in}\, \Sigma_c \quad .. \quad L_n : \tau_n''\, \mathbf{in}\, D(\tau_1, .., \tau_i)\, \mathbf{in}\, \Sigma_c \\ \Sigma_c;\Gamma, x_1{:}_{\mathit{eff}'}\tau_1'' \vdash_{\mathit{eff}_0} ce_1 \Leftarrow \tau \quad .. \quad \Sigma_c;\Gamma, x_n{:}_{\mathit{eff}'}\tau_n'' \vdash_{\mathit{eff}_0} ce_n \Leftarrow \tau\end{array}}{\Sigma_c;\Gamma \vdash_{\mathit{eff}_0} \mathbf{case}\, ce\, \mathbf{of}\, \{L_1\, x_1 \to ce_1 | .. | L_n\, x_n \to ce_n\} \Leftarrow \tau} \quad \textsc{Chk_case}$$

$$\frac{\begin{array}{l}\mathsf{type}\ D(a, b) = \{\mathsf{Next}{:}a \mid \mathsf{Done}{:}b\} \in \Sigma_c \\ \Sigma_c;\Gamma \vdash_{\mathbf{0}} ce_1 \Rightarrow A \\ \Sigma_c;\Gamma, x{:}_{\mathbf{0}} A \vdash_{\mathbf{1}} ce_2 \Leftarrow D(A, B)\end{array}}{\Sigma_c;\Gamma \vdash_{\mathbf{1}} \mathbf{iter}\, (x = ce_1)\{ce_2\} \Leftarrow B} \quad \textsc{Chk_iter}$$

$$\frac{\begin{array}{l}\mathit{eff}' = \lfloor \mathit{eff}_0 \rfloor \\ \Sigma_c;\Gamma \vdash_{\mathit{eff}'} ce_1 \Leftarrow \mathsf{Bool} \\ \Sigma_c;\Gamma \vdash_{\mathit{eff}_0} ce_2 \Leftarrow \tau \\ \Sigma_c;\Gamma \vdash_{\mathit{eff}_0} ce_3 \Leftarrow \tau\end{array}}{\Sigma_c;\Gamma \vdash_{\mathit{eff}_0} \mathbf{if}\, ce_1\, \mathbf{then}\, ce_2\, \mathbf{else}\, ce_3 \Leftarrow \tau} \quad \textsc{Chk_if}$$

$$\frac{\Sigma_c;\Gamma \vdash_{\mathbf{S}} ce \Leftarrow \tau}{\Sigma_c;\Gamma \vdash_{\mathbf{S}} \mathbf{return}\, ce \Leftarrow \mathsf{Pred}\ \tau} \quad \textsc{Chk_return}$$

$$\frac{\begin{array}{l}\Sigma_c;\Gamma \vdash_{\mathbf{S}} ce_1 \Rightarrow \mathsf{Pred}\ \tau \\ \Sigma_c;\Gamma, x{:}_{\mathbf{S}}\tau \vdash_{\mathbf{S}} ce_2 \Leftarrow \mathsf{Pred}\ \tau'\end{array}}{\Sigma_c;\Gamma \vdash_{\mathbf{S}} \mathbf{take}\, x = ce_1; ce_2 \Leftarrow \mathsf{Pred}\ \tau'} \quad \textsc{Chk_take}$$

$$\frac{\Sigma_c;\Gamma \vdash_{\mathit{eff}_0} ce \Rightarrow \tau}{\Sigma_c;\Gamma \vdash_{\mathit{eff}_0} ce \Leftarrow \tau} \quad \textsc{Chk_sub}$$

$\boxed{\Sigma_c \vdash \mathit{cprog}}$

$$\frac{\Sigma_c;\cdot \vdash_{\mathit{eff}} ce \Leftarrow \tau}{\Sigma_c \vdash \mathsf{main} : \tau_{\mathit{eff}} = ce} \quad \textsc{CProg_main}$$

$$\frac{\begin{array}{l}\Sigma_c;\cdot, x{:}_{\mathbf{0}} A \vdash_{\mathbf{0}} ce \Leftarrow B \\ \Sigma_c, \mathsf{fun}\ f(x : A) \to_{\mathbf{0}} B = ce \vdash \mathit{cprog}\end{array}}{\Sigma_c \vdash \mathsf{fun}\ f(x : A) \to_{\mathbf{0}} B = ce \quad \mathit{cprog}} \quad \textsc{CProg_purefun}$$

$$\frac{\begin{array}{l}\Sigma_c, f : A \to_{\mathbf{1}} B; \cdot, x{:}_{\mathbf{0}} A \vdash_{\mathbf{1}} ce \Leftarrow B \\ \Sigma_c, f : A \to_{\mathbf{1}} B \vdash \mathit{cprog}\end{array}}{\Sigma_c \vdash \mathsf{fun}\ f(x : A) \to_{\mathbf{1}} B = ce \quad \mathit{cprog}} \quad \textsc{CProg_impurefun}$$

$$\frac{\Sigma_c, f : \tau \to_{\mathbf{S}} \tau'; \cdot, x{:}_{\mathbf{S}}\tau \vdash_{\mathbf{S}} ce \Leftarrow \tau' \quad \Sigma_c, \mathsf{fun}\, f(x:\tau) \to_{\mathbf{S}} \tau' = ce \vdash cprog}{\Sigma_c \vdash \mathsf{fun}\, f(x:\tau) \to_{\mathbf{S}} \tau' = ce \quad cprog} \quad \textsc{CProg_specfun}$$

$$\frac{\Sigma_c \vdash \mathsf{sort}\ D(\overline{a}) = \{\overline{L:\tau}\}\ \mathsf{ok} \quad \Sigma_c, \mathsf{sort}\ D(\overline{a}) = \{\overline{L:\tau}\} \vdash cprog}{\Sigma_c \vdash \mathbf{sort}\, D(a_1, .., a_i) = \{L_1 : \tau_1 | .. | L_n : \tau_n\} cprog} \quad \textsc{CProg_sortdecl}$$

$$\frac{\Sigma_c \vdash \mathsf{type}\ D(\overline{a}) = \{\overline{L:A}\}\ \mathsf{ok} \quad \Sigma_c, \mathsf{type}\ D(\overline{a}) = \{\overline{L:A}\} \vdash cprog}{\Sigma_c \vdash \mathbf{type}\, D(a_1, .., a_i) = \{L_1 : A_1 | .. | L_n : A_n\} cprog} \quad \textsc{CProg_typedecl}$$

$\boxed{L : \tau\, \mathbf{in}\, D(\tau_1, .., \tau_i)\, \mathbf{in}\, \Sigma_c}$

$$\frac{\mathbf{sort}\, D(a_1, .., a_i)\, \mathbf{in}\, \Sigma_c \quad L{:}\tau' \in \mathrm{ctors}(D) \in \Sigma_c \quad \tau = [\cdot, \tau_1/a_1, .., \tau_i/a_i]\tau'}{L : \tau\, \mathbf{in}\, D(\tau_1, .., \tau_i)\, \mathbf{in}\, \Sigma_c} \quad \textsc{Ctor_sort}$$

$$\frac{\mathbf{type}\, D(a_1, .., a_i)\, \mathbf{in}\, \Sigma_c \quad L{:}A' \in \mathsf{tyctors}(D) \in \Sigma_c \quad A = [\cdot, A_1/a_1, .., A_i/a_i]A'}{L : A\, \mathbf{in}\, D(A_1, .., A_i)\, \mathbf{in}\, \Sigma_c} \quad \textsc{Ctor_type}$$

$\boxed{D(a_1, .., a_k) = \{L_1 : \tau_1 | .. | L_n : \tau_n\} \in \Sigma_c}$

$$\frac{\mathbf{sort}\, D(a_1, .., a_k) = \{L_1 : \tau_1 | .. | L_n : \tau_n\}\, \mathbf{in}\, \Sigma_c}{D(a_1, .., a_k) = \{L_1 : \tau_1 | .. | L_n : \tau_n\} \in \Sigma_c} \quad \textsc{DL_sort}$$

$$\frac{\mathbf{type}\, D(a_1, .., a_k) = \{L_1 : A_1 | .. | L_n : A_n\}\, \mathbf{in}\, \Sigma_c}{D(a_1, .., a_k) = \{L_1 : A_1 | .. | L_n : A_n\} \in \Sigma_c} \quad \textsc{DL_type}$$

$\boxed{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} se \Rightarrow \tau \rightsquigarrow ce}$

$$\frac{x{:}_{\mathit{eff}'}\tau \in \Gamma \quad \mathit{eff}' \leq \mathit{eff}_0}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} x \Rightarrow \tau \rightsquigarrow x} \quad \textsc{SurfSyn_var}$$

$$\frac{}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathit{lit} \Rightarrow \mathsf{Int} \rightsquigarrow \mathit{lit}} \quad \textsc{SurfSyn_int}$$

$$\frac{}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathit{blit} \Rightarrow \mathsf{Bool} \rightsquigarrow \mathit{blit}} \quad \textsc{SurfSyn_bool}$$

$$\frac{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} se \Leftarrow \tau \rightsquigarrow ce}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} se : \tau \Rightarrow \tau \rightsquigarrow ce : \tau} \quad \textsc{SurfSyn_annot}$$

$$\frac{\begin{array}{l} \mathit{prim}{:}\tau \to_{\mathit{eff}'} \tau' \\ \mathit{eff}' \leq \mathit{eff}_0 \\ \mathit{eff}'' = \lfloor \mathit{eff}_0 \rfloor \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}''} se \Leftarrow \tau \rightsquigarrow ce \end{array}}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathit{prim}\ se \Rightarrow \tau' \rightsquigarrow \mathit{prim}\ ce} \quad \textsc{SurfSyn_app}$$

$$\frac{\begin{array}{l} f{:}\tau \to_{\mathit{eff}'} \tau' \in \Sigma_c \\ \mathit{eff}' \leq \mathit{eff}_0 \\ \mathit{eff}'' = \lfloor \mathit{eff}_0 \rfloor \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}''} se \Leftarrow \tau \rightsquigarrow ce \end{array}}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} f\ se \Rightarrow \tau' \rightsquigarrow f\ ce} \quad \textsc{SurfSyn_call}$$

$$\frac{\begin{array}{l} A\ \mathsf{eqtype} \\ \mathit{eff}' = \lfloor \mathit{eff}_0 \rfloor \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}'} se_1 \Rightarrow A \rightsquigarrow ce_1 \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}'} se_2 \Leftarrow A \rightsquigarrow ce_2 \end{array}}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} se_1 \doteq se_2 \Rightarrow \mathsf{Bool} \rightsquigarrow ce_1 \doteq ce_2} \quad \textsc{SurfSyn_eq}$$

$\boxed{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} se \Leftarrow \tau \rightsquigarrow ce}$

$$\frac{\begin{array}{l}\Sigma_c;\Gamma \vdash_{\mathit{eff}_0} se_1 \Rightarrow \tau \rightsquigarrow ce_1\\ \epsilon = \lfloor \mathit{eff}_0 \rfloor\\ \Sigma_c;\Gamma \vdash_{\mathit{eff}_0} y,\cdot;pat:\tau \to [\cdot][[\cdot]]se_2 \Leftarrow_\epsilon \tau' \rightsquigarrow ce_2\end{array}}{\Sigma_c;\Gamma \vdash_{\mathit{eff}_0} \mathbf{let}\, pat = se_1; se_2 \Leftarrow \tau' \rightsquigarrow \mathbf{let}\, y = ce_1; ce_2} \quad \textsc{SurfChk_let}$$

$$\frac{\Sigma_c;\Gamma \vdash_{\mathit{eff}_0} se_1 \Leftarrow \tau_1 \rightsquigarrow ce_1 \quad .. \quad \Sigma_c;\Gamma \vdash_{\mathit{eff}_0} se_n \Leftarrow \tau_n \rightsquigarrow ce_n}{\Sigma_c;\Gamma \vdash_{\mathit{eff}_0} (se_1, .., se_n) \Leftarrow (\tau_1, .., \tau_n) \rightsquigarrow (ce_1, .., ce_n)} \quad \textsc{SurfChk_tuple}$$

$$\frac{\begin{array}{l}L:\tau\,\mathbf{in}\,D(\tau_1, .., \tau_i)\,\mathbf{in}\,\Sigma_c\\ \mathit{eff}' = \lfloor \mathit{eff}_0 \rfloor\\ \Sigma_c;\Gamma \vdash_{\mathit{eff}'} se \Leftarrow \tau \rightsquigarrow ce\end{array}}{\Sigma_c;\Gamma \vdash_{\mathit{eff}_0} L\,se \Leftarrow D(\tau_1, .., \tau_i) \rightsquigarrow L\,ce} \quad \textsc{SurfChk_inject}$$

$$\frac{\begin{array}{l}\epsilon = \lfloor \mathit{eff}_0 \rfloor\\ \Sigma_c;\Gamma \vdash_\epsilon se \Rightarrow \tau \rightsquigarrow ce'\\ M = \overline{pat{:}\tau, \cdot \to [\cdot][\cdot]\,se}\\ \Sigma_c;\Gamma \vdash_{\mathit{eff}_0} y,\cdot;M \Leftarrow_\epsilon \tau' \rightsquigarrow ce\end{array}}{\Sigma_c;\Gamma \vdash_{\mathit{eff}_0} \mathbf{case}\,se\,\mathbf{of}\,\{pat_1 \to se_1|..|pat_n \to se_n\} \Leftarrow \tau' \rightsquigarrow \mathbf{let}\, y = ce'; ce} \quad \textsc{SurfChk_case}$$

$$\frac{\begin{array}{l}\Sigma_c;\Gamma \vdash_{\mathbf{0}} se_1 \Rightarrow A \rightsquigarrow ce_1\\ \mathsf{type}\ D(a,b) = \{\mathsf{Next}{:}a \mid \mathsf{Done}{:}b\} \in \Sigma_c\\ \Sigma_c;\Gamma \vdash_{\mathbf{1}} z,\cdot;pat:A,\cdot \to [\cdot][[\cdot]]se_2 \Leftarrow_{\mathbf{0}} D(A,B) \rightsquigarrow ce_2\end{array}}{\Sigma_c;\Gamma \vdash_{\mathbf{1}} \mathbf{iter}\,(pat = se_1)\{se_2\} \Leftarrow B \rightsquigarrow \mathbf{iter}\,(z = ce_1)\{ce_2\}} \quad \textsc{SurfChk_iter}$$

$$\frac{\begin{array}{l}\mathit{eff}' = \lfloor \mathit{eff}_0 \rfloor\\ \Sigma_c;\Gamma \vdash_{\mathit{eff}'} se_1 \Leftarrow \mathsf{Bool} \rightsquigarrow ce_1\\ \Sigma_c;\Gamma \vdash_{\mathit{eff}_0} se_2 \Leftarrow \tau \rightsquigarrow ce_2\\ \Sigma_c;\Gamma \vdash_{\mathit{eff}_0} se_3 \Leftarrow \tau \rightsquigarrow ce_3\end{array}}{\Sigma_c;\Gamma \vdash_{\mathit{eff}_0} \mathbf{if}\,se_1\,\mathbf{then}\,se_2\,\mathbf{else}\,se_3 \Leftarrow \tau \rightsquigarrow \mathbf{if}\,ce_1\,\mathbf{then}\,ce_2\,\mathbf{else}\,ce_3} \quad \textsc{SurfChk_if}$$

$$\frac{\Sigma_c;\Gamma \vdash_{\mathbf{S}} se \Leftarrow \tau \rightsquigarrow ce}{\Sigma_c;\Gamma \vdash_{\mathbf{S}} \mathbf{return}\,se \Leftarrow \mathsf{Pred}\ \tau \rightsquigarrow \mathbf{return}\,ce} \quad \textsc{SurfChk_return}$$

$$\frac{\begin{array}{l}\Sigma_c;\Gamma \vdash_{\mathbf{S}} se_1 \Rightarrow \mathsf{Pred}\ \tau \rightsquigarrow ce_1\\ \Sigma_c;\Gamma \vdash_{\mathbf{S}} y,\cdot;pat:\tau \to [\cdot][[\cdot]]se_2 \Leftarrow_{\mathbf{S}} \mathsf{Pred}\ \tau' \rightsquigarrow ce_2\end{array}}{\Sigma_c;\Gamma \vdash_{\mathbf{S}} \mathbf{take}\,pat = se_1; se_2 \Leftarrow \mathsf{Pred}\ \tau' \rightsquigarrow \mathbf{take}\,y = ce_1; ce_2} \quad \textsc{SurfChk_take}$$

$$\frac{\Sigma_c;\Gamma \vdash_{\mathit{eff}_0} se \Rightarrow \tau \rightsquigarrow ce}{\Sigma_c;\Gamma \vdash_{\mathit{eff}_0} se \Leftarrow \tau \rightsquigarrow ce} \quad \textsc{SurfChk_sub}$$

$$\boxed{\begin{array}{l}\Sigma_s \vdash prog\\ \quad \rightsquigarrow \Sigma_c \vdash cprog\end{array}}$$

$$\frac{\Sigma_c;\cdot \vdash_{\mathit{eff}} se \Leftarrow \tau \rightsquigarrow ce}{\begin{array}{l}\Sigma_s \vdash \mathsf{main} : \tau_{\mathit{eff}} = se\\ \quad \rightsquigarrow \Sigma_c \vdash \mathsf{main} : \tau_{\mathit{eff}} = ce\end{array}} \quad \textsc{Prog_main}$$

$$\frac{\begin{array}{l}M = \overline{pat{:}A, \cdot \to [\cdot][\cdot]\,se}\\ \Sigma_c;\cdot \vdash_{\mathbf{0}} y,\cdot;M \Leftarrow_{\mathbf{0}} B \rightsquigarrow ce\\ \Sigma_s, \mathsf{fun}\ f : A \to_{\mathbf{0}} B = \{\overline{\mathbf{pat} \to \mathbf{se}}\} \vdash prog\\ \quad \rightsquigarrow \Sigma_c, \mathsf{fun}\ f(y:A) \to_{\mathbf{0}} B = ce \vdash cprog\end{array}}{\begin{array}{l}\Sigma_s \vdash \mathsf{fun}\ f : A \to_{\mathbf{0}} B = \{\overline{\mathbf{pat} \to \mathbf{se}}\} \quad prog\\ \quad \rightsquigarrow \Sigma_c \vdash \mathsf{fun}\ f(y:A) \to_{\mathbf{0}} B = ce \quad cprog\end{array}} \quad \textsc{Prog_purefun}$$

$$\frac{\begin{array}{l}M = \overline{pat{:}A, \cdot \to [\cdot][\cdot]\,se}\\ \Sigma_c, f : A \to_{\mathbf{1}} B;\cdot \vdash_{\mathbf{1}} y,\cdot;M \Leftarrow_{\mathbf{0}} B \rightsquigarrow ce\\ \Sigma_s, f : A \to_{\mathbf{1}} B \vdash prog\\ \quad \rightsquigarrow \Sigma_c, f : A \to_{\mathbf{1}} B \vdash cprog\end{array}}{\begin{array}{l}\Sigma_s \vdash \mathsf{fun}\ f : A \to_{\mathbf{1}} B = \{\overline{\mathbf{pat} \to \mathbf{se}}\} \quad prog\\ \quad \rightsquigarrow \Sigma_c \vdash \mathsf{fun}\ f(y:A) \to_{\mathbf{1}} B = ce \quad cprog\end{array}} \quad \textsc{Prog_impurefun}$$

$$\frac{\begin{array}{l} M = \overline{pat{:}\tau, \cdot \to [\cdot][\cdot]\, se} \\ \Sigma_c, f : \tau \to_{\mathbf{S}} \tau'; \cdot \vdash_{\mathbf{S}} y, \cdot; M \Leftarrow_{\mathbf{S}} \tau' \rightsquigarrow ce \\ \Sigma_s, \mathsf{fun}\ f : \tau \to_{\mathbf{S}} \tau' = \{\overline{\mathbf{pat} \to \mathbf{se}}\} \vdash prog \\ \quad \rightsquigarrow \Sigma_c, \mathsf{fun}\ f(y : \tau) \to_{\mathbf{S}} \tau' = ce \vdash cprog \end{array}}{\begin{array}{l} \Sigma_s \vdash \mathsf{fun}\ f : \tau \to_{\mathbf{S}} \tau' = \{\overline{\mathbf{pat} \to \mathbf{se}}\} \quad prog \\ \quad \rightsquigarrow \Sigma_c \vdash \mathsf{fun}\ f(y : \tau) \to_{\mathbf{S}} \tau' = ce \quad cprog \end{array}} \quad \textsc{Prog_specfun}$$

$$\frac{\begin{array}{l} \Sigma_c \vdash \mathsf{sort}\ D(\overline{a}) = \{\overline{L : \tau}\}\ \mathsf{ok} \\ \Sigma_s, \mathsf{sort}\ D(\overline{a}) = \{\overline{L : \tau}\} \vdash prog \\ \quad \rightsquigarrow \Sigma_c, \mathsf{sort}\ D(\overline{a}) = \{\overline{L : \tau}\} \vdash cprog \end{array}}{\begin{array}{l} \Sigma_s \vdash \mathbf{sort}\, D(a_1, .., a_i) = \{L_1 : \tau_1 | .. | L_n : \tau_n\} prog \\ \quad \rightsquigarrow \Sigma_c \vdash \mathbf{sort}\, D(a_1, .., a_i) = \{L_1 : \tau_1 | .. | L_n : \tau_n\} cprog \end{array}} \quad \textsc{Prog_sortdecl}$$

$$\frac{\begin{array}{l} \Sigma_c \vdash \mathsf{type}\ D(\overline{a}) = \{\overline{L : A}\}\ \mathsf{ok} \\ \Sigma_s, \mathsf{type}\ D(\overline{a}) = \{\overline{L : A}\} \vdash prog \\ \quad \rightsquigarrow \Sigma_c, \mathsf{type}\ D(\overline{a}) = \{\overline{L : A}\} \vdash cprog \end{array}}{\begin{array}{l} \Sigma_s \vdash \mathbf{type}\, D(a_1, .., a_i) = \{L_1 : A_1 | .. | L_n : A_n\} prog \\ \quad \rightsquigarrow \Sigma_c \vdash \mathbf{type}\, D(a_1, .., a_i) = \{L_1 : A_1 | .. | L_n : A_n\} cprog \end{array}} \quad \textsc{Prog_typedecl}$$

$\boxed{M\ \mathsf{has_con}}$

$$\frac{}{L\, pat : \tau, \beta \to [\Gamma][\mathcal{E}]se \mid M\ \mathsf{has_con}} \quad \text{HC_HERE}$$

$$\frac{M\ \mathsf{has_con}}{\beta \to [\Gamma][\mathcal{E}]se \mid M\ \mathsf{has_con}} \quad \text{HC_THERE}$$

$\boxed{M\ \mathsf{has_tup}}$

$$\frac{}{(pat_1, .., pat_n) : \tau, \beta \to [\Gamma][\mathcal{E}]se \mid M\ \mathsf{has_tup}} \quad \text{HT_HERE}$$

$$\frac{M\ \mathsf{has_tup}}{\beta \to [\Gamma][\mathcal{E}]se \mid M\ \mathsf{has_tup}} \quad \text{HT_THERE}$$

$\boxed{M/var \overset{y}{\rightsquigarrow}_{\mathit{eff}} M'}$

$$\frac{}{\cdot/var \overset{y}{\rightsquigarrow}_{\mathit{eff}} \cdot} \quad \text{SV_EMPTY}$$

$$\frac{M/var \overset{y}{\rightsquigarrow}_{\mathit{eff}} M'}{x : \tau, \beta \to [\Gamma][\mathcal{E}]se \mid M/var \overset{y}{\rightsquigarrow}_{\mathit{eff}} \beta \to [\Gamma, x{:}_{\mathit{eff}}\tau][\mathbf{let}\, x = y; \mathcal{E}]se \mid M'} \quad \text{SV_CONS}$$

$\boxed{M/L{:}\tau \overset{y}{\rightsquigarrow}_{\mathit{eff}} M'}$

$$\frac{}{\cdot/L{:}\tau \overset{y}{\rightsquigarrow}_{\mathit{eff}} \cdot} \quad \text{SC_EMPTY}$$

$$\frac{M/L{:}\tau \overset{y}{\rightsquigarrow}_{\mathit{eff}} M'}{L\, pat : \tau', \beta \to [\Gamma][\mathcal{E}]se \mid M/L{:}\tau \overset{y}{\rightsquigarrow}_{\mathit{eff}} pat : \tau, \beta \to [\Gamma][\mathcal{E}]se \mid M'} \quad \text{SC_MATCH}$$

$$\frac{L \neq L' \quad M/L{:}\tau \overset{y}{\rightsquigarrow}_{\mathit{eff}} M'}{L'\, pat : \tau', \beta \to [\Gamma][\mathcal{E}]se \mid M/L{:}\tau \overset{y}{\rightsquigarrow}_{\mathit{eff}} M'} \quad \text{SC_SKIP}$$

$$\frac{M/L{:}\tau \overset{y}{\rightsquigarrow}_{\mathit{eff}} M'}{x : \tau', \beta \to [\Gamma][\mathcal{E}]se \mid M/L{:}\tau \overset{y}{\rightsquigarrow}_{\mathit{eff}} z : \tau, \beta \to [\Gamma, x{:}_{\mathit{eff}}\tau'][\mathbf{let}\, x = y; \mathcal{E}]se \mid M'} \quad \text{SC_VAR}$$

$\boxed{M/\mathbf{tup} \overset{y}{\rightsquigarrow}_{\mathit{eff}} M'}$

$$\frac{}{\cdot/\mathbf{tup} \overset{y}{\rightsquigarrow}_{\mathit{eff}} \cdot} \quad \text{ET_EMPTY}$$

$$\frac{M/\mathbf{tup} \overset{y}{\rightsquigarrow}_{\mathit{eff}} M'}{(pat_1, .., pat_k) : (\tau_1, .., \tau_k), \beta \to [\Gamma][\mathcal{E}]se \mid M/\mathbf{tup} \overset{y}{\rightsquigarrow}_{\mathit{eff}} pat_1 : \tau_1, .., pat_k : \tau_k, \beta \to [\Gamma][\mathcal{E}]se \mid M'} \quad \text{ET_TUPLE}$$

$$\frac{M/\mathbf{tup} \overset{y}{\rightsquigarrow}_{\mathit{eff}} M'}{x : (\tau_1, .., \tau_k), \beta \to [\Gamma][\mathcal{E}]se \mid M/\mathbf{tup} \overset{y}{\rightsquigarrow}_{\mathit{eff}} z_1 : \tau_1, .., z_k : \tau_k, \beta \to [\Gamma, x{:}_{\mathit{eff}}(\tau_1, .., \tau_k)][\mathbf{let}\, x = y; \mathcal{E}]se \mid M'} \quad \text{ET_VAR}$$

$\boxed{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \overline{y}; M \Leftarrow_\epsilon \tau \rightsquigarrow ce}$

$$\frac{\Sigma_c; \Gamma, \Gamma' \vdash_{\mathit{eff}_0} se \Leftarrow \tau \rightsquigarrow ce' \quad ce = \mathcal{E}[ce']}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \cdot; \cdot \to [\Gamma'][\mathcal{E}]se \mid M \Leftarrow_\epsilon \tau \rightsquigarrow ce} \quad \text{Cov_DONE}$$

$$\frac{\begin{array}{l} M/var \overset{y}{\rightsquigarrow}_\epsilon M' \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \overline{y}; M' \Leftarrow_\epsilon \tau \rightsquigarrow ce \end{array}}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} y, \overline{y}; M \Leftarrow_\epsilon \tau \rightsquigarrow ce} \quad \text{Cov_VAR}$$

$$\frac{\begin{array}{l} M \ \mathsf{has_con} \\ D(a_1, .., a_k) = \{L_1 : \tau_1' | .. | L_n : \tau_n'\} \in \Sigma_c \\ L_1 : \tau_1'' \,\mathbf{in}\, D(\tau_1, .., \tau_i) \,\mathbf{in}\, \Sigma_c \quad .. \quad L_n : \tau_n'' \,\mathbf{in}\, D(\tau_1, .., \tau_i) \,\mathbf{in}\, \Sigma_c \\ M/L_1{:}\tau_1'' \overset{y}{\rightsquigarrow}_\epsilon M_1 \quad .. \quad M/L_n{:}\tau_n'' \overset{y}{\rightsquigarrow}_\epsilon M_n \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}_0} x_1, \overline{y}; M_1 \Leftarrow_\epsilon \tau \rightsquigarrow ce_1 \quad .. \quad \Sigma_c; \Gamma \vdash_{\mathit{eff}_0} x_n, \overline{y}; M_n \Leftarrow_\epsilon \tau \rightsquigarrow ce_n \end{array}}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} y, \overline{y}; M \Leftarrow_\epsilon \tau \rightsquigarrow \mathbf{case}\, y \,\mathbf{of}\, \{L_1\, x_1 \to ce_1 | .. | L_n\, x_n \to ce_n\}} \quad \text{Cov_CON}$$

$$\frac{\begin{array}{l} M \ \mathsf{has_tup} \\ M/\mathbf{tup} \overset{y}{\rightsquigarrow}_\epsilon M' \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}_0} y_1, .., y_n, \overline{y}; M' \Leftarrow_\epsilon \tau \rightsquigarrow ce \end{array}}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} y, \overline{y}; M \Leftarrow_\epsilon \tau \rightsquigarrow \mathbf{let}\, (y_1, .., y_n) = y; ce} \quad \text{Cov_TUP}$$

$\boxed{\mathcal{E}[ce] \rightsquigarrow ce'}$

$$\frac{}{[\cdot][ce] \rightsquigarrow ce} \quad \text{Fill_HOLE}$$

$$\frac{\mathcal{E}[ce] \rightsquigarrow ce'}{\mathbf{let}\, x = ce''; \mathcal{E}[ce] \rightsquigarrow \mathbf{let}\, x = ce''; ce'} \quad \text{Fill_LET}$$

$\boxed{\Sigma_r \vdash \Delta \ \mathsf{wf}}$

$$\frac{}{\Sigma_r \vdash \cdot \ \mathsf{wf}} \quad \text{RWf_EMPTY}$$

$$\frac{\Sigma_r \vdash \Delta \ \mathsf{wf} \quad \mathit{eff} \leq \mathbf{S}}{\Sigma_r \vdash \Delta, x{:}_{\mathit{eff}}\tau \ \mathsf{wf}} \quad \text{RWf_EFF}$$

$$\frac{\Sigma_r \vdash \Delta \ \mathsf{wf} \quad \Gamma = |\Delta| \quad \Sigma_c = \mathsf{Comp}(\Sigma_r) \quad \Sigma_c; \Gamma \vdash_{\mathbf{S}} ce \Leftarrow \mathsf{Bool}}{\Sigma_r \vdash \Delta, x{:}ce \ [\mathsf{log}] \ \mathsf{wf}} \quad \text{RWf_LOG}$$

$$\frac{\Sigma_r \vdash \Delta \ \mathsf{wf} \quad \Gamma = |\Delta| \quad \Sigma_c = \mathsf{Comp}(\Sigma_r) \quad \Sigma_c; \Gamma \vdash_{\mathbf{S}} ce \Rightarrow \mathsf{Pred}\ \tau \quad \Sigma_c; \Gamma \vdash_{\mathbf{S}} ce' \Leftarrow \tau}{\Sigma_r \vdash \Delta, x{:}ce@ce' \ [\mathsf{res}(u)] \ \mathsf{wf}} \quad \text{RWf_RES}$$

$\boxed{\Gamma = |\Delta|}$

$$\frac{}{\cdot = |\cdot|} \quad \text{Erase_EMPTY}$$

$$\frac{\Gamma = |\Delta|}{\Gamma, x{:}_{\mathit{eff}}\tau = |\Delta, x{:}_{\mathit{eff}}\tau|} \quad \text{Erase_EFF}$$

$$\frac{\Gamma = |\Delta|}{\Gamma = |\Delta, x{:}ce\ [\mathsf{log}]|}\ \textsc{Erase_log}$$

$$\frac{\Gamma = |\Delta|}{\Gamma = |\Delta, x{:}ce@ce'\ [\mathsf{res}(u)]|}\ \textsc{Erase_res}$$

$\boxed{\Delta' = \mathsf{affinize}(\Delta)}$

$$\frac{}{\cdot = \mathsf{affinize}(\cdot)}\ \textsc{Aff_empty}$$

$$\frac{\Delta' = \mathsf{affinize}(\Delta)}{\Delta', x{:}_{\mathit{eff}}\tau = \mathsf{affinize}(\Delta, x{:}_{\mathit{eff}}\tau)}\ \textsc{Aff_eff}$$

$$\frac{\Delta' = \mathsf{affinize}(\Delta)}{\Delta', x{:}ce\ [\mathsf{log}] = \mathsf{affinize}(\Delta, x{:}ce\ [\mathsf{log}])}\ \textsc{Aff_log}$$

$$\frac{\Delta' = \mathsf{affinize}(\Delta)}{\Delta', x{:}ce@ce'\ [\mathsf{res}(0)] = \mathsf{affinize}(\Delta, x{:}ce@ce'\ [\mathsf{res}(0)])}\ \textsc{Aff_res_used}$$

$$\frac{\Delta' = \mathsf{affinize}(\Delta)}{\Delta', x{:}ce@ce'\ [\mathsf{res}(?)] = \mathsf{affinize}(\Delta, x{:}ce@ce'\ [\mathsf{res}(1)])}\ \textsc{Aff_res_avail}$$

$$\frac{\Delta' = \mathsf{affinize}(\Delta)}{\Delta', x{:}ce@ce'\ [\mathsf{res}(?)] = \mathsf{affinize}(\Delta, x{:}ce@ce'\ [\mathsf{res}(?)])}\ \textsc{Aff_res_opt}$$

$\boxed{\mathsf{zero}(\Delta)}$

$$\frac{}{\mathsf{zero}(\cdot)}\ \textsc{Zero_empty}$$

$$\frac{\mathsf{zero}(\Delta)}{\mathsf{zero}(\Delta, x{:}_{\mathit{eff}}\tau)}\ \textsc{Zero_eff}$$

$$\frac{\mathsf{zero}(\Delta)}{\mathsf{zero}(\Delta, x{:}ce\ [\mathsf{log}])}\ \textsc{Zero_log}$$

$$\frac{\mathsf{zero}(\Delta) \quad u \in \{0, ?\}}{\mathsf{zero}(\Delta, x{:}ce@ce'\ [\mathsf{res}(u)])}\ \textsc{Zero_res}$$

$\boxed{\mathcal{C} = \Delta \Rightarrow \mathcal{C}'}$

$$\frac{}{\mathcal{C} = \cdot \Rightarrow \mathcal{C}}\ \textsc{Close_empty}$$

$$\frac{\mathcal{C} = \Delta \Rightarrow \forall x{:}\tau.\, \mathcal{C}'}{\mathcal{C} = \Delta, x{:}_{\mathit{eff}}\tau \Rightarrow \mathcal{C}'}\ \textsc{Close_eff}$$

$$\frac{\mathcal{C} = \Delta \Rightarrow ce \Rightarrow \mathcal{C}'}{\mathcal{C} = \Delta, x{:}ce\ [\mathsf{log}] \Rightarrow \mathcal{C}'}\ \textsc{Close_log}$$

$$\frac{\mathcal{C} = \Delta \Rightarrow \mathcal{C}'}{\mathcal{C} = \Delta, x{:}ce@ce'\ [\mathsf{res}(u)] \Rightarrow \mathcal{C}'}\ \textsc{Close_res}$$

$\boxed{\Sigma_r; \Gamma \vdash \mathcal{C}\ \mathsf{wf}}$

$$\frac{}{\Sigma_r; \Gamma \vdash \top\ \mathsf{wf}}\ \textsc{CtWf_top}$$

$$\frac{\Sigma_r; \Gamma \vdash \mathcal{C}_1\ \mathsf{wf} \quad \Sigma_r; \Gamma \vdash \mathcal{C}_2\ \mathsf{wf}}{\Sigma_r; \Gamma \vdash \mathcal{C}_1 \wedge \mathcal{C}_2\ \mathsf{wf}}\ \textsc{CtWf_and}$$

$$\frac{\Sigma_r; \Gamma, x{:}_{\mathbf{S}}\tau \vdash \mathcal{C}\ \mathsf{wf}}{\Sigma_r; \Gamma \vdash \forall x{:}\tau.\, \mathcal{C}\ \mathsf{wf}}\ \textsc{CtWf_forall}$$

$$\frac{\Sigma_c = \mathsf{Comp}(\Sigma_r) \quad \Sigma_c; \Gamma \vdash_{\mathbf{S}} ce \Leftarrow \mathsf{Bool} \quad \Sigma_r; \Gamma \vdash \mathcal{C}\ \mathsf{wf}}{\Sigma_r; \Gamma \vdash ce \Rightarrow \mathcal{C}\ \mathsf{wf}} \quad \textsc{CtWf_impl}$$

$$\frac{\Sigma_c = \mathsf{Comp}(\Sigma_r) \quad \Sigma_c; \Gamma \vdash_{\mathbf{S}} ce \Leftarrow \mathsf{Bool}}{\Sigma_r; \Gamma \vdash ce\ \mathsf{wf}} \quad \textsc{CtWf_atom}$$

$$\frac{}{\Sigma_r; \Gamma \vdash \bot\ \mathsf{wf}} \quad \textsc{CtWf_bot}$$

$\boxed{\Sigma_r; \Gamma \vdash_{\mathit{eff}} Pf\ \mathsf{wf}}$

$$\frac{}{\Sigma_r; \Gamma \vdash_{\mathit{eff}} \cdot\ \mathsf{wf}} \quad \textsc{PfWf_nil}$$

$$\frac{\begin{array}{c}\mathit{eff}' \in \{\mathbf{S}, \lfloor \mathit{eff} \rfloor\} \\ \Sigma_r; \Gamma, x{:}_{\mathit{eff}'}\tau \vdash_{\mathit{eff}} Pf\ \mathsf{wf}\end{array}}{\Sigma_r; \Gamma \vdash_{\mathit{eff}} x{:}_{\mathit{eff}'}\tau, Pf\ \mathsf{wf}} \quad \textsc{PfWf_comp}$$

$$\frac{\Sigma_c = \mathsf{Comp}(\Sigma_r) \quad \Sigma_c; \Gamma \vdash_{\mathbf{S}} ce \Leftarrow \mathsf{Bool} \quad \Sigma_r; \Gamma \vdash_{\mathit{eff}} Pf\ \mathsf{wf}}{\Sigma_r; \Gamma \vdash_{\mathit{eff}} ce\ [\mathsf{log}], Pf\ \mathsf{wf}} \quad \textsc{PfWf_log}$$

$$\frac{\Sigma_c = \mathsf{Comp}(\Sigma_r) \quad \Sigma_c; \Gamma \vdash_{\mathbf{S}} ce' \Rightarrow \tau \quad \Sigma_c; \Gamma \vdash_{\mathbf{S}} ce \Leftarrow \mathsf{Pred}\ \tau \quad \Sigma_r; \Gamma \vdash_{\mathit{eff}} Pf\ \mathsf{wf}}{\Sigma_r; \Gamma \vdash_{\mathit{eff}} ce@ce'\ [\mathsf{res}], Pf\ \mathsf{wf}} \quad \textsc{PfWf_res}$$

$$\frac{\Sigma_c = \mathsf{Comp}(\Sigma_r) \quad \Sigma_c; \Gamma \vdash_{\mathbf{S}} ce \Rightarrow \mathsf{Pred}\ \tau \quad \Sigma_r; \Gamma, y{:}_{\mathbf{S}}\tau \vdash_{\mathit{eff}} Pf\ \mathsf{wf}}{\Sigma_r; \Gamma \vdash_{\mathit{eff}} (y).ce\ [\mathsf{res}], Pf\ \mathsf{wf}} \quad \textsc{PfWf_depres}$$

$\boxed{\Sigma_r \vdash RF\ \mathsf{wf}}$

$$\frac{\Sigma_r; \cdot \vdash_{\mathit{eff}} Pf_1\ \mathsf{wf} \quad \Gamma = \mathsf{Bind}(\cdot; Pf_1) \quad \Sigma_r; \Gamma \vdash_{\mathit{eff}} Pf_2\ \mathsf{wf}}{\Sigma_r \vdash Pf_1 \multimap Pf_2\ [\mathit{eff}]\ \mathsf{wf}} \quad \textsc{RFWf_rfun}$$

$\boxed{\Sigma_r \vdash f{:}RF}$

$$\frac{f{:}RF \in \Sigma_r}{\Sigma_r \vdash f{:}RF} \quad \textsc{RSLk_rfun}$$

$$\frac{f{:}\tau \to_{\mathit{eff}} \tau' \in \Sigma_r}{\Sigma_r \vdash f{:}x{:}_{\mathit{eff}}\tau, \cdot \multimap y{:}_{\mathit{eff}}\tau', \cdot\ [\mathit{eff}]} \quad \textsc{RSLk_sig}$$

$$\frac{f(\mathbf{x} : \tau) \to_{\mathit{eff}} \tau' = ce \in \Sigma_r}{\Sigma_r \vdash f{:}x{:}_{\mathit{eff}}\tau, \cdot \multimap y{:}_{\mathit{eff}}\tau', \cdot\ [\mathit{eff}]} \quad \textsc{RSLk_def}$$

$\boxed{\overline{A} = \{Pf\}}$

$$\frac{}{[] = \{\cdot\}} \quad \textsc{PfTys_nil}$$

$$\frac{\overline{A} = \{Pf\}}{A{::}\overline{A} = \{x{:}_{\mathbf{0}}A, Pf\}} \quad \textsc{PfTys_pure}$$

$$\frac{\overline{A} = \{Pf\}}{\overline{A} = \{x{:}_{\mathbf{S}}\tau, Pf\}} \quad \textsc{PfTys_spec}$$

$$\frac{\overline{A} = \{Pf\}}{\overline{A} = \{ce\ [\mathsf{log}], Pf\}} \quad \textsc{PfTys_log}$$

$$\frac{\overline{A} = \{Pf\}}{\overline{A} = \{ce@ce'\ [\mathsf{res}], Pf\}} \quad \textsc{PfTys_res}$$

$$\frac{\overline{A} = \{Pf\}}{\overline{A} = \{(y).ce\ [\mathsf{res}], Pf\}} \quad \textsc{PfTys_depres}$$

$\boxed{\Sigma_c = \mathsf{Comp}(\Sigma_r)}$

$$\frac{}{\cdot = \mathsf{Comp}(\cdot)} \quad \textsc{RSErase_empty}$$

$$\frac{\Sigma_c = \mathsf{Comp}(\Sigma_r)}{\Sigma_c, f : F = \mathsf{Comp}(\Sigma_r, f : F)} \quad \textsc{RSErase_ext}$$

$$\frac{\Sigma_c = \mathsf{Comp}(\Sigma_r)}{\Sigma_c, \mathsf{fun}\ f(x : \tau) \to_{\mathit{eff}} \tau' = ce = \mathsf{Comp}(\Sigma_r, \mathsf{fun}\ f(x : \tau) \to_{\mathit{eff}} \tau' = ce)} \quad \textsc{RSErase_fundef}$$

$$\frac{\Sigma_c = \mathsf{Comp}(\Sigma_r) \quad \overline{A}_1 = \{Pf_1\} \quad A = \mathsf{Prod}(\overline{A}_1) \quad \overline{A}_2 = \{Pf_2\} \quad B = \mathsf{Prod}(\overline{A}_2)}{\Sigma_c, f : A \to_{\mathit{eff}} B = \mathsf{Comp}(\Sigma_r, f : Pf_1 \multimap Pf_2\ [\mathit{eff}])} \quad \textsc{RSErase_rfun}$$

$$\frac{\Sigma_c = \mathsf{Comp}(\Sigma_r)}{\Sigma_c, \mathsf{sort}\ D(\overline{a}) = \{\overline{L : \tau}\} = \mathsf{Comp}(\Sigma_r, \mathsf{sort}\ D(\overline{a}) = \{\overline{L : \tau}\})} \quad \textsc{RSErase_sort}$$

$$\frac{\Sigma_c = \mathsf{Comp}(\Sigma_r)}{\Sigma_c, \mathsf{type}\ D(\overline{a}) = \{\overline{L : A}\} = \mathsf{Comp}(\Sigma_r, \mathsf{type}\ D(\overline{a}) = \{\overline{L : A}\})} \quad \textsc{RSErase_type}$$

$\boxed{\vdash \Sigma_r\ \mathsf{wf}}$

$$\frac{}{\vdash \cdot\ \mathsf{wf}} \quad \textsc{RSWf_empty}$$

$$\frac{\vdash \Sigma_r\ \mathsf{wf}}{\vdash \Sigma_r, f : F\ \mathsf{wf}} \quad \textsc{RSWf_ext}$$

$$\frac{\vdash \Sigma_r\ \mathsf{wf} \quad \Sigma_c = \mathsf{Comp}(\Sigma_r) \quad \Sigma_c; \Gamma, x{:}_{\mathit{eff}}\tau \vdash_{\mathit{eff}} ce \Leftarrow \tau'}{\vdash \Sigma_r, \mathsf{fun}\ f(x : \tau) \to_{\mathit{eff}} \tau' = ce\ \mathsf{wf}} \quad \textsc{RSWf_fundef}$$

$$\frac{\vdash \Sigma_r\ \mathsf{wf} \quad \Sigma_c = \mathsf{Comp}(\Sigma_r) \quad \Sigma_r \vdash RF\ \mathsf{wf}}{\vdash \Sigma_r, f : RF\ \mathsf{wf}} \quad \textsc{RSWf_rfun}$$

$\boxed{\Sigma_r; \Delta \vdash \mathit{lpf} \Rightarrow ce \dashv \Delta' \rightsquigarrow \mathcal{C}}$

$$\frac{x{:}ce\ [\mathsf{log}] \in \Delta}{\Sigma_r; \Delta \vdash x \Rightarrow ce \dashv \Delta \rightsquigarrow \top} \quad \textsc{LSyn_var}$$

$$\frac{\begin{array}{l}\Sigma_c = \mathsf{Comp}(\Sigma_r) \quad f(\mathbf{x} : \tau) \to_{\mathit{eff}} \tau' = ce' \in \Sigma_c \quad \mathit{eff} \leq \mathbf{S} \\ \Gamma = |\Delta| \quad \Sigma_c; \Gamma \vdash_{\mathbf{S}} ce \Leftarrow \tau\end{array}}{\Sigma_r; \Delta \vdash \mathsf{unfold}\ f(ce) \Rightarrow (f\ ce) \doteq [\mathsf{id}(\Gamma), ce/x]ce' \dashv \Delta \rightsquigarrow \top} \quad \textsc{LSyn_unfold}$$

$$\frac{\Sigma_r; \Delta \vdash \mathit{lpf} \Leftarrow ce \dashv \Delta' \rightsquigarrow \mathcal{C}}{\Sigma_r; \Delta \vdash \mathit{lpf} : ce \Rightarrow ce \dashv \Delta' \rightsquigarrow \mathcal{C}} \quad \textsc{LSyn_annot}$$

$$\frac{\Sigma_r; \Delta \vdash \mathit{rpf} \Rightarrow \mathbf{return}\ ce@ce' \dashv \Delta' \rightsquigarrow \mathcal{C}}{\Sigma_r; \Delta \vdash \mathsf{open\text{-}ret}\ \mathit{rpf} \Rightarrow ce \doteq ce' \dashv \Delta' \rightsquigarrow \mathcal{C}} \quad \textsc{LSyn_openret}$$

$\boxed{\Sigma_r; \Delta \vdash \mathit{lpf} \Leftarrow ce \dashv \Delta' \rightsquigarrow \mathcal{C}}$

$$\frac{}{\Sigma_r; \Delta \vdash \mathsf{auto} \Leftarrow ce \dashv \Delta \rightsquigarrow ce} \quad \textsc{LChk_auto}$$

$$\frac{\Sigma_r; \Delta \vdash \mathit{lpf} \Rightarrow ce \dashv \Delta' \rightsquigarrow \mathcal{C}}{\Sigma_r; \Delta \vdash \mathit{lpf} \Leftarrow ce' \dashv \Delta' \rightsquigarrow \mathcal{C} \wedge (ce \Rightarrow (ce'))} \quad \textsc{LChk_sub}$$

$\boxed{\Sigma_r; \Delta \vdash x : ce@ce' \dashv \Delta'}$

$$\frac{x \neq y \quad \Sigma_r; \Delta \vdash x : ce@ce' \dashv \Delta'}{\Sigma_r; \Delta, y{:}_{\mathit{eff}}\tau \vdash x : ce@ce' \dashv \Delta', y{:}_{\mathit{eff}}\tau} \quad \textsc{RLk_there_eff}$$

$$\frac{x \neq y \quad \Sigma_r; \Delta \vdash x : ce@ce' \dashv \Delta'}{\Sigma_r; \Delta, y{:}ce''\ [\mathsf{log}] \vdash x : ce@ce' \dashv \Delta', y{:}ce''\ [\mathsf{log}]} \quad \textsc{RLk_there_log}$$

$$\frac{x \neq y \quad \Sigma_r; \Delta \vdash x : ce@ce' \dashv \Delta'}{\Sigma_r; \Delta, y{:}ce''@ce''' \; [\mathsf{res}(u)] \vdash x : ce@ce' \dashv \Delta', y{:}ce''@ce''' \; [\mathsf{res}(u)]} \quad \textsc{RLk_there_res}$$

$$\frac{u \in \{1, ?\}}{\Sigma_r; \Delta, x{:}ce@ce' \; [\mathsf{res}(u)] \vdash x : ce@ce' \dashv \Delta, x{:}ce@ce' \; [\mathsf{res}(0)]} \quad \textsc{RLk_here}$$

$\boxed{\Sigma_r; \Delta \vdash rpf \Rightarrow ce@ce' \dashv \Delta' \rightsquigarrow \mathcal{C}}$

$$\frac{\Sigma_r; \Delta \vdash x : ce@ce' \dashv \Delta'}{\Sigma_r; \Delta \vdash x \Rightarrow ce@ce' \dashv \Delta' \rightsquigarrow \top} \quad \textsc{RSyn_var}$$

$$\frac{\Sigma_r; \Delta \vdash rpf \Leftarrow ce@ce' \dashv \Delta' \rightsquigarrow \mathcal{C}}{\Sigma_r; \Delta \vdash rpf : ce@ce' \Rightarrow ce@ce' \dashv \Delta' \rightsquigarrow \mathcal{C}} \quad \textsc{RSyn_annot}$$

$\boxed{\Sigma_r; \Delta \vdash rpf \Leftarrow ce@ce' \dashv \Delta' \rightsquigarrow \mathcal{C}}$

$$\frac{\Sigma_r; \Delta \vdash lpf \Leftarrow ce_1 \doteq ce_2 \dashv \Delta' \rightsquigarrow \mathcal{C}}{\Sigma_r; \Delta \vdash \mathsf{make\text{-}ret}\ lpf \Leftarrow \mathbf{return}\ ce_1@ce_2 \dashv \Delta' \rightsquigarrow \mathcal{C}} \quad \textsc{RChk_makeret}$$

$$\frac{\Sigma_r; \Delta \vdash_{\mathit{eff}} crt \Leftarrow x{:}_{\mathbf{S}}\tau, ce_1@x \; [\mathsf{res}], ce_2@ce_3 \; [\mathsf{res}], \cdot \dashv \Delta' \rightsquigarrow \mathcal{C}}{\Sigma_r; \Delta \vdash \mathsf{make\text{-}take}\ crt \Leftarrow \mathbf{take}\, x = ce_1; ce_2@ce_3 \dashv \Delta' \rightsquigarrow \mathcal{C}} \quad \textsc{RChk_maketake}$$

$$\frac{\Sigma_r; \Delta \vdash rpf \Rightarrow ce_1@ce_1' \dashv \Delta' \rightsquigarrow \mathcal{C}}{\Sigma_r; \Delta \vdash rpf \Leftarrow ce_2@ce_2' \dashv \Delta' \rightsquigarrow (\mathcal{C} \wedge (ce_1 \doteq ce_2)) \wedge (ce_1' \doteq ce_2')} \quad \textsc{RChk_sub}$$

$\boxed{\Sigma_r; \Delta \vdash Pf_1 = Pf_2 \rightsquigarrow \mathcal{C}}$

$$\frac{}{\Sigma_r; \Delta \vdash \cdot = \cdot \rightsquigarrow \top} \quad \textsc{PfEq_nil}$$

$$\frac{\Sigma_r; \Delta, x{:}_{\mathit{eff}}\tau \vdash Pf_1 = [\mathsf{id}(|\Delta|), x/y]Pf_2 \rightsquigarrow \mathcal{C}}{\Sigma_r; \Delta \vdash x{:}_{\mathit{eff}}\tau, Pf_1 = y{:}_{\mathit{eff}}\tau, Pf_2 \rightsquigarrow \forall x{:}\tau.\, \mathcal{C}} \quad \textsc{PfEq_comp}$$

$$\frac{\Sigma_r; \Delta \vdash Pf_1 = Pf_2 \rightsquigarrow \mathcal{C}}{\Sigma_r; \Delta \vdash ce_1 \; [\mathsf{log}], Pf_1 = ce_2 \; [\mathsf{log}], Pf_2 \rightsquigarrow ((ce_1 \doteq ce_2)) \wedge \mathcal{C}} \quad \textsc{PfEq_log}$$

$$\frac{\Sigma_r; \Delta \vdash Pf_1 = Pf_2 \rightsquigarrow \mathcal{C}}{\Sigma_r; \Delta \vdash ce_1@ce_1' \; [\mathsf{res}], Pf_1 = ce_2@ce_2' \; [\mathsf{res}], Pf_2 \rightsquigarrow (((ce_1 \doteq ce_2)) \wedge (ce_1' \doteq ce_2')) \wedge \mathcal{C}} \quad \textsc{PfEq_res}$$

$$\frac{\begin{array}{c}\Gamma = |\Delta| \quad \Sigma_c = \mathsf{Comp}(\Sigma_r) \quad \Sigma_c; \Gamma \vdash_{\mathbf{S}} ce_1 \Rightarrow \mathsf{Pred}\ \tau \quad \Sigma_c; \Gamma \vdash_{\mathbf{S}} ce_2 \Rightarrow \mathsf{Pred}\ \tau \\ \Sigma_r; \Delta, z{:}_{\mathbf{S}}\tau \vdash [\mathsf{id}(|\Delta|), z/y]Pf_1 = [\mathsf{id}(|\Delta|), z/w]Pf_2 \rightsquigarrow \mathcal{C}\end{array}}{\Sigma_r; \Delta \vdash (y).ce_1 \; [\mathsf{res}], Pf_1 = (w).ce_2 \; [\mathsf{res}], Pf_2 \rightsquigarrow ((ce_1 \doteq ce_2)) \wedge \forall z{:}\tau.\, \mathcal{C}} \quad \textsc{PfEq_depres}$$

$\boxed{\Sigma_r \vdash \Delta_0 \sqcap \Delta_1 = \Delta}$

$$\frac{}{\Sigma_r \vdash \cdot \sqcap \cdot = \cdot} \quad \textsc{Merge_empty}$$

$$\frac{\Sigma_r \vdash \Delta_0 \sqcap \Delta_1 = \Delta}{\Sigma_r \vdash \Delta_0, x{:}_{\mathit{eff}}\tau \sqcap \Delta_1, x{:}_{\mathit{eff}}\tau = \Delta, x{:}_{\mathit{eff}}\tau} \quad \textsc{Merge_eff}$$

$$\frac{\Sigma_r \vdash \Delta_0 \sqcap \Delta_1 = \Delta}{\Sigma_r \vdash \Delta_0, x{:}ce \; [\mathsf{log}] \sqcap \Delta_1, x{:}ce \; [\mathsf{log}] = \Delta, x{:}ce \; [\mathsf{log}]} \quad \textsc{Merge_log}$$

$$\frac{\Sigma_r \vdash \Delta_0 \sqcap \Delta_1 = \Delta \quad u = u_0 \sqcap u_1}{\Sigma_r \vdash \Delta_0, x{:}ce@ce' \; [\mathsf{res}(u_0)] \sqcap \Delta_1, x{:}ce@ce' \; [\mathsf{res}(u_1)] = \Delta, x{:}ce@ce' \; [\mathsf{res}(u)]} \quad \textsc{Merge_res}$$

$\boxed{\Sigma_r; \Gamma \vdash q : Pf \dashv \Delta}$

$$\frac{}{\Sigma_r; \Gamma \vdash () : \cdot \dashv \cdot} \quad \textsc{RPM_empty}$$

$$\frac{\Sigma_r; \Gamma, x{:}_{\mathit{eff}}\tau \vdash q : [\mathsf{id}(\Gamma), x/y]Pf \dashv \Delta}{\Sigma_r; \Gamma \vdash (x, q) : y{:}_{\mathit{eff}}\tau, Pf \dashv \cdot, x{:}_{\mathit{eff}}\tau, \Delta} \quad \textsc{RPM_comp}$$

$$\frac{\Sigma_r; \Gamma \vdash q : Pf \dashv \Delta}{\Sigma_r; \Gamma \vdash (x, q) : ce\ [\mathsf{log}], Pf \dashv \cdot, x{:}ce\ [\mathsf{log}], \Delta} \quad \text{RPM_LOG}$$

$$\frac{\Sigma_r; \Gamma \vdash q : Pf \dashv \Delta}{\Sigma_r; \Gamma \vdash (x, q) : ce@ce'\ [\mathsf{res}], Pf \dashv \cdot, x{:}ce@ce'\ [\mathsf{res}(1)], \Delta} \quad \text{RPM_RES}$$

$$\frac{\begin{array}{c}\Sigma_c = \mathsf{Comp}(\Sigma_r) \quad \Sigma_c; \Gamma \vdash_{\mathbf{S}} ce \Rightarrow \mathsf{Pred}\ \tau \\ \Sigma_r; \Gamma, x{:}_{\mathbf{S}}\tau \vdash q : [\mathsf{id}(\Gamma), x/z]Pf \dashv \Delta\end{array}}{\Sigma_r; \Gamma \vdash ((x, w), q) : (z).ce\ [\mathsf{res}], Pf \dashv \cdot, x{:}_{\mathbf{S}}\tau, w{:}ce@x\ [\mathsf{res}(1)], \Delta} \quad \text{RPM_DEPRES}$$

$\boxed{\Sigma_r; \Delta \vdash_{\mathit{eff}} crt \Rightarrow Pf \dashv \Delta' \rightsquigarrow \mathcal{C}}$

$$\frac{\begin{array}{c}\Gamma = |\Delta| \quad \Sigma_r; \Gamma \vdash_{\mathit{eff}} Pf\ \mathsf{wf} \\ \Sigma_r; \Delta \vdash_{\mathit{eff}} crt \Leftarrow Pf \dashv \Delta' \rightsquigarrow \mathcal{C}\end{array}}{\Sigma_r; \Delta \vdash_{\mathit{eff}} crt : Pf \Rightarrow Pf \dashv \Delta' \rightsquigarrow \mathcal{C}} \quad \text{RCSyn_annot}$$

$$\frac{\begin{array}{c}\Sigma_r \vdash f{:}Pf_1 \multimap Pf_2\ [\mathit{eff}'] \quad \mathit{eff}' \leq \mathit{eff} \quad \mathit{eff}'' = \lfloor \mathit{eff} \rfloor \\ \Sigma_r; \Delta \vdash_{\mathit{eff}''} rsp : Pf_1 \multimap Pf_2 \gg Pf \dashv \Delta' \rightsquigarrow \mathcal{C}\end{array}}{\Sigma_r; \Delta \vdash_{\mathit{eff}} f\ rsp \Rightarrow Pf \dashv \Delta' \rightsquigarrow \mathcal{C}} \quad \text{RCSyn_call}$$

$$\frac{\begin{array}{c}prim{:}Pf_1 \multimap Pf_2\ [\mathit{eff}'] \quad \mathit{eff}' \leq \mathit{eff} \quad \mathit{eff}'' = \lfloor \mathit{eff} \rfloor \\ \Sigma_r; \Delta \vdash_{\mathit{eff}''} rsp : Pf_1 \multimap Pf_2 \gg Pf \dashv \Delta' \rightsquigarrow \mathcal{C}\end{array}}{\Sigma_r; \Delta \vdash_{\mathit{eff}} prim\ rsp \Rightarrow Pf \dashv \Delta' \rightsquigarrow \mathcal{C}} \quad \text{RCSyn_primapp}$$

$$\frac{\Sigma_r; \Delta \vdash rpf \Rightarrow \mathbf{take}\, x = ce_1; ce_2@ce_3 \dashv \Delta' \rightsquigarrow \mathcal{C}}{\Sigma_r; \Delta \vdash_{\mathit{eff}} \mathsf{open\text{-}take}\ rpf \Rightarrow x{:}_{\mathbf{S}}\tau, ce_1@x\ [\mathsf{res}], ce_2@ce_3\ [\mathsf{res}], \cdot \dashv \Delta' \rightsquigarrow \mathcal{C}} \quad \text{RCSyn_opentake}$$

$$\frac{\begin{array}{l}\Sigma_c = \mathsf{Comp}(\Sigma_r) \quad \mathsf{type}\ D(a, b) = \{\mathsf{Next}{:}a \mid \mathsf{Done}{:}b\} \in \Sigma_c \\ \Gamma = |\Delta| \quad \Sigma_c; \Gamma \vdash_{\mathbf{S}} ce \Rightarrow \mathsf{Pred}\ D(A, B) \\ \Sigma_r; \Delta \vdash_{\mathbf{0}} crt_1 \Leftarrow x{:}_{\mathbf{0}}A, ce@Next\ x\ [\mathsf{res}], \cdot \dashv \Delta' \rightsquigarrow \mathcal{C} \\ \Sigma_r; \Delta', x{:}_{\mathbf{0}}A, y{:}ce@Next\ x\ [\mathsf{res}(1)] \vdash_{\mathbf{1}} crt_2 \Leftarrow z{:}_{\mathbf{0}}D(A, B), ce@z\ [\mathsf{res}], \cdot \dashv \Delta', x{:}_{\mathbf{0}}A, y{:}ce@Next\ x\ [\mathsf{res}(0)] \rightsquigarrow \mathcal{C}'\end{array}}{\Sigma_r; \Delta \vdash_{\mathbf{1}} \mathbf{iter}\, [ce](q = crt_1)\{crt_2\} \Rightarrow z{:}_{\mathbf{0}}B, ce@Done\ z\ [\mathsf{res}], \cdot \dashv \Delta' \rightsquigarrow \mathcal{C} \wedge \forall x{:}A.\, \mathcal{C}'} \quad \text{RCSy}$$

$\boxed{\Sigma_r; \Delta \vdash_{\mathit{eff}} y{:}ce\ \mathsf{is}\ L\, x \rightarrow crt \Leftarrow Pf \dashv \Delta' \rightsquigarrow \mathcal{C}}$

$$\frac{\begin{array}{l}\Gamma = |\Delta| \quad \Sigma_c = \mathsf{Comp}(\Sigma_r) \quad \Sigma_c; \Gamma \vdash_{\mathit{eff}} ce \Rightarrow D(\tau_1, \ldots, \tau_i) \\ L : \tau\ \mathbf{in}\ D(\tau_1, \ldots, \tau_i)\ \mathbf{in}\ \Sigma_c \\ \Sigma_r; \Delta, x{:}_{\mathit{eff}}\tau, y{:}ce \doteq L\, x\ [\mathsf{log}] \vdash_{\mathit{eff}} crt \Leftarrow Pf \dashv \Delta', x{:}_{\mathit{eff}}\tau, y{:}ce \doteq L\, x\ [\mathsf{log}] \rightsquigarrow \mathcal{C}\end{array}}{\Sigma_r; \Delta \vdash_{\mathit{eff}} y{:}ce\ \mathsf{is}\ L\, x \rightarrow crt \Leftarrow Pf \dashv \Delta' \rightsquigarrow \forall x{:}\tau.\, ce \doteq L\, x \Rightarrow \mathcal{C}} \quad \text{CBr_branch}$$

$\boxed{\Sigma_r; \Delta \vdash_{\mathit{eff}} crt \Leftarrow Pf \dashv \Delta' \rightsquigarrow \mathcal{C}}$

$$\frac{\begin{array}{l}\Sigma_r; \Delta \vdash_{\mathit{eff}} crt_1 \Rightarrow Pf' \dashv \Delta' \rightsquigarrow \mathcal{C}_1 \\ \Gamma = |\Delta'| \quad \Sigma_r; \Gamma \vdash q : Pf' \dashv \Delta'' \\ \Sigma_r; \Delta', \Delta'' \vdash_{\mathit{eff}} crt_2 \Leftarrow Pf \dashv \Delta''', \Delta'''' \rightsquigarrow \mathcal{C}_2 \\ \mathsf{len}(\Delta') = \mathsf{len}(\Delta''') \\ \mathsf{len}(\Delta'') = \mathsf{len}(\Delta'''') \\ \mathsf{zero}(\Delta'''')\end{array}}{\Sigma_r; \Delta \vdash_{\mathit{eff}} \mathbf{let}\ q = crt_1; crt_2 \Leftarrow Pf \dashv \Delta''' \rightsquigarrow \mathcal{C}_1 \wedge \mathcal{C}_2} \quad \text{RCChk_let}$$

$$\frac{\begin{array}{l}\mathit{eff}' = \lfloor \mathit{eff} \rfloor \\ \Gamma = |\Delta| \quad \Sigma_c = \mathsf{Comp}(\Sigma_r) \quad \Sigma_c; \Gamma \vdash_{\mathit{eff}'} ce \Leftarrow \mathsf{Bool} \\ \Sigma_r; \Delta, x{:}ce \doteq \mathsf{true}\ [\mathsf{log}] \vdash_{\mathit{eff}'} crt_1 \Leftarrow Pf \dashv \Delta_1, x{:}ce \doteq \mathsf{true}\ [\mathsf{log}] \rightsquigarrow \mathcal{C}_1 \\ \Sigma_r; \Delta, x{:}ce \doteq \mathsf{false}\ [\mathsf{log}] \vdash_{\mathit{eff}'} crt_2 \Leftarrow Pf \dashv \Delta_2, x{:}ce \doteq \mathsf{false}\ [\mathsf{log}] \rightsquigarrow \mathcal{C}_2 \\ \Sigma_r \vdash \Delta_1 \sqcap \Delta_2 = \Delta'\end{array}}{\Sigma_r; \Delta \vdash_{\mathit{eff}} \mathbf{if}\ [x]ce\ \mathbf{then}\ crt_1\ \mathbf{else}\ crt_2 \Leftarrow Pf \dashv \Delta' \rightsquigarrow (ce \doteq \mathsf{true} \Rightarrow \mathcal{C}_1) \wedge (ce \doteq \mathsf{false} \Rightarrow \mathcal{C}_2)} \quad \text{RCChk_if}$$

$$\frac{\begin{array}{l} \mathit{eff}' = \lfloor \mathit{eff} \rfloor \\ \Gamma = |\Delta| \quad \Sigma_c = \mathsf{Comp}(\Sigma_r) \quad \Sigma_c; \Gamma \vdash_{\mathit{eff}'} ce \Rightarrow D(\tau_1, .., \tau_i) \\ D(a_1, .., a_k) = \{L_1 : \tau_1' | .. | L_n : \tau_n'\} \in \Sigma_c \\ \Sigma_r; \Delta \vdash_{\mathit{eff}'} y{:}ce \text{ is } L_1\, x_1 \to crt_1 \Leftarrow Pf \dashv \Delta_1 \rightsquigarrow \mathcal{C}_1 \quad .. \quad \Sigma_r; \Delta \vdash_{\mathit{eff}'} y{:}ce \text{ is } L_n\, x_n \to crt_n \Leftarrow Pf \dashv \Delta_n \rightsquigarrow \mathcal{C}_n \\ \Sigma_r \vdash \sqcap(\Delta_1; ..; \Delta_n) = \Delta' \end{array}}{\Sigma_r; \Delta \vdash_{\mathit{eff}} \mathbf{case}\, [y] ce\, \mathbf{of}\, \{L_1\, x_1 \to crt_1 | .. | L_n\, x_n \to crt_n\} \Leftarrow Pf \dashv \Delta' \rightsquigarrow \mathcal{C}_1 \wedge \mathcal{C}_n} \quad \textsc{RCChk}$$

$$\frac{\Delta' = \mathsf{affinize}(\Delta)}{\Sigma_r; \Delta \vdash_{\mathit{eff}} \mathsf{exfalso} \Leftarrow Pf \dashv \Delta' \rightsquigarrow \bot} \quad \textsc{RCChk_exfalso}$$

$$\frac{\begin{array}{l} \Sigma_r; \Delta \vdash_{\mathit{eff}} crt \Rightarrow Pf' \dashv \Delta' \rightsquigarrow \mathcal{C} \\ \Sigma_r; \Delta' \vdash Pf' = Pf \rightsquigarrow \mathcal{C}' \end{array}}{\Sigma_r; \Delta \vdash_{\mathit{eff}} crt \Leftarrow Pf \dashv \Delta' \rightsquigarrow \mathcal{C} \wedge \mathcal{C}'} \quad \textsc{RCChk_sub}$$

$\boxed{\Sigma_r; \Delta \vdash_{\mathit{eff}} rsp : Pf_1 \multimap Pf_2 \gg Pf \dashv \Delta' \rightsquigarrow \mathcal{C}}$

$$\frac{}{\Sigma_r; \Delta \vdash_{\mathit{eff}} \cdot : \cdot \multimap Pf \gg Pf \dashv \Delta \rightsquigarrow \top} \quad \textsc{Spine_nil}$$

$$\frac{\begin{array}{l} \Gamma = |\Delta| \quad \Sigma_c = \mathsf{Comp}(\Sigma_r) \quad \Sigma_c; \Gamma \vdash_{\mathit{eff}} ce \Leftarrow \tau \\ \Sigma_r; \Delta \vdash_{\mathit{eff}} rsp : [\mathsf{id}(|\Delta|), ce/x]Pf_1 \multimap [\mathsf{id}(|\Delta|), ce/x]Pf_2 \gg Pf \dashv \Delta' \rightsquigarrow \mathcal{C} \end{array}}{\Sigma_r; \Delta \vdash_{\mathit{eff}} ce, rsp : x{:}_{\mathbf{0}}\tau, Pf_1 \multimap Pf_2 \gg Pf \dashv \Delta' \rightsquigarrow \mathcal{C}} \quad \textsc{Spine_pure}$$

$$\frac{\begin{array}{l} \Gamma = |\Delta| \quad \Sigma_c = \mathsf{Comp}(\Sigma_r) \quad \Sigma_c; \Gamma \vdash_{\mathbf{S}} ce \Leftarrow \tau \\ \Sigma_r; \Delta \vdash_{\mathit{eff}} rsp : [\mathsf{id}(|\Delta|), ce/x]Pf_1 \multimap [\mathsf{id}(|\Delta|), ce/x]Pf_2 \gg Pf \dashv \Delta' \rightsquigarrow \mathcal{C} \end{array}}{\Sigma_r; \Delta \vdash_{\mathit{eff}} ce, rsp : x{:}_{\mathbf{S}}\tau, Pf_1 \multimap Pf_2 \gg Pf \dashv \Delta' \rightsquigarrow \mathcal{C}} \quad \textsc{Spine_spec}$$

$$\frac{\begin{array}{l} \Sigma_r; \Delta \vdash lpf \Leftarrow ce \dashv \Delta' \rightsquigarrow \mathcal{C} \\ \Sigma_r; \Delta' \vdash_{\mathit{eff}} rsp : Pf_1 \multimap Pf_2 \gg Pf \dashv \Delta'' \rightsquigarrow \mathcal{C}' \end{array}}{\Sigma_r; \Delta \vdash_{\mathit{eff}} lpf, rsp : ce\ [\mathsf{log}], Pf_1 \multimap Pf_2 \gg Pf \dashv \Delta'' \rightsquigarrow \mathcal{C} \wedge \mathcal{C}'} \quad \textsc{Spine_log}$$

$$\frac{\begin{array}{l} \Sigma_r; \Delta \vdash rpf \Leftarrow ce_1 @ ce_2 \dashv \Delta' \rightsquigarrow \mathcal{C} \\ \Sigma_r; \Delta' \vdash_{\mathit{eff}} rsp : Pf_1 \multimap Pf_2 \gg Pf \dashv \Delta'' \rightsquigarrow \mathcal{C}' \end{array}}{\Sigma_r; \Delta \vdash_{\mathit{eff}} rpf, rsp : ce_1 @ ce_2\ [\mathsf{res}], Pf_1 \multimap Pf_2 \gg Pf \dashv \Delta'' \rightsquigarrow \mathcal{C} \wedge \mathcal{C}'} \quad \textsc{Spine_res}$$

$$\frac{\begin{array}{l} \Sigma_r; \Delta \vdash rpf \Rightarrow ce' @ ce'' \dashv \Delta' \rightsquigarrow \mathcal{C} \\ \Sigma_r; \Delta' \vdash_{\mathit{eff}} rsp : [\mathsf{id}(|\Delta'|), ce''/y]Pf_1 \multimap [\mathsf{id}(|\Delta'|), ce''/y]Pf_2 \gg Pf \dashv \Delta'' \rightsquigarrow \mathcal{C}' \end{array}}{\Sigma_r; \Delta \vdash_{\mathit{eff}} rpf, rsp : (y).ce\ [\mathsf{res}], Pf_1 \multimap Pf_2 \gg Pf \dashv \Delta'' \rightsquigarrow (\mathcal{C} \wedge (ce \doteq ce')) \wedge \mathcal{C}'} \quad \textsc{Spine_depres}$$

$\boxed{\Sigma_r; \Delta \vdash_{\mathit{eff}} rsp : Pf \dashv \Delta' \rightsquigarrow \mathcal{C}}$

$$\frac{}{\Sigma_r; \Delta \vdash_{\mathit{eff}} \cdot : \cdot \dashv \Delta \rightsquigarrow \top} \quad \textsc{Tup_nil}$$

$$\frac{\begin{array}{l} \Gamma = |\Delta| \quad \Sigma_c = \mathsf{Comp}(\Sigma_r) \quad \Sigma_c; \Gamma \vdash_{\mathit{eff}} ce \Leftarrow \tau \\ \Sigma_r; \Delta \vdash_{\mathit{eff}} rsp : [\mathsf{id}(|\Delta|), ce/x]Pf \dashv \Delta' \rightsquigarrow \mathcal{C} \end{array}}{\Sigma_r; \Delta \vdash_{\mathit{eff}} ce, rsp : x{:}_{\mathit{eff}}\tau, Pf \dashv \Delta' \rightsquigarrow \mathcal{C}} \quad \textsc{Tup_core}$$

$$\frac{\begin{array}{l} \Sigma_r; \Delta \vdash lpf \Leftarrow ce \dashv \Delta' \rightsquigarrow \mathcal{C} \\ \Sigma_r; \Delta' \vdash_{\mathit{eff}} rsp : Pf \dashv \Delta'' \rightsquigarrow \mathcal{C}' \end{array}}{\Sigma_r; \Delta \vdash_{\mathit{eff}} lpf, rsp : ce\ [\mathsf{log}], Pf \dashv \Delta'' \rightsquigarrow \mathcal{C} \wedge \mathcal{C}'} \quad \textsc{Tup_log}$$

$$\frac{\begin{array}{l} \Sigma_r; \Delta \vdash rpf \Leftarrow ce @ ce' \dashv \Delta' \rightsquigarrow \mathcal{C} \\ \Sigma_r; \Delta' \vdash_{\mathit{eff}} rsp : Pf \dashv \Delta'' \rightsquigarrow \mathcal{C}' \end{array}}{\Sigma_r; \Delta \vdash_{\mathit{eff}} rpf, rsp : ce @ ce'\ [\mathsf{res}], Pf \dashv \Delta'' \rightsquigarrow \mathcal{C} \wedge \mathcal{C}'} \quad \textsc{Tup_res}$$

$$\frac{\begin{array}{l} \Sigma_r; \Delta \vdash rpf \Rightarrow ce' @ ce'' \dashv \Delta' \rightsquigarrow \mathcal{C} \\ \Sigma_r; \Delta' \vdash_{\mathit{eff}} rsp : [\mathsf{id}(|\Delta'|), ce''/y]Pf \dashv \Delta'' \rightsquigarrow \mathcal{C}' \end{array}}{\Sigma_r; \Delta \vdash_{\mathit{eff}} rpf, rsp : (y).ce\ [\mathsf{res}], Pf \dashv \Delta'' \rightsquigarrow (\mathcal{C} \wedge (ce \doteq ce')) \wedge \mathcal{C}'} \quad \textsc{Tup_depres}$$

$\boxed{\Sigma_r \vdash rprog \dashv \Sigma_r' \rightsquigarrow \mathcal{C}}$

$$\frac{\Sigma_r; \cdot \vdash_{\mathit{eff}} \mathit{crt} \Leftarrow \mathit{Pf} \dashv \Delta' \rightsquigarrow \mathcal{C}}{\Sigma_r \vdash \mathsf{main} : \mathit{Pf}_{\mathit{eff}} = \mathit{crt} \dashv \Sigma_r \rightsquigarrow \mathcal{C}} \quad \text{RProg_main}$$

$$\frac{\begin{array}{c}\Sigma_c = \mathsf{Comp}(\Sigma_r) \quad \Sigma_c \vdash \mathsf{sort}\ D(\overline{a}) = \{\overline{L : \tau}\}\ \mathsf{ok} \\ \Sigma_r, \mathsf{sort}\ D(\overline{a}) = \{\overline{L : \tau}\} \vdash \mathit{rprog} \dashv \Sigma_r' \rightsquigarrow \mathcal{C}\end{array}}{\Sigma_r \vdash \mathbf{sort}\, D(a_1, .., a_i) = \{L_1 : \tau_1 | .. | L_n : \tau_n\} \mathit{rprog} \dashv \Sigma_r' \rightsquigarrow \mathcal{C}} \quad \text{RProg_sortdecl}$$

$$\frac{\begin{array}{c}\Sigma_c = \mathsf{Comp}(\Sigma_r) \quad \Sigma_c \vdash \mathsf{type}\ D(\overline{a}) = \{\overline{L : A}\}\ \mathsf{ok} \\ \Sigma_r, \mathsf{type}\ D(\overline{a}) = \{\overline{L : A}\} \vdash \mathit{rprog} \dashv \Sigma_r' \rightsquigarrow \mathcal{C}\end{array}}{\Sigma_r \vdash \mathbf{type}\, D(a_1, .., a_i) = \{L_1 : A_1 | .. | L_n : A_n\} \mathit{rprog} \dashv \Sigma_r' \rightsquigarrow \mathcal{C}} \quad \text{RProg_typedecl}$$

$$\frac{\begin{array}{c}\Sigma_c = \mathsf{Comp}(\Sigma_r) \quad \Sigma_c; \cdot, x{:}_{\mathbf{0}} A \vdash_{\mathbf{0}} \mathit{ce} \Leftarrow B \\ \Sigma_r, \mathsf{fun}\ f(x : A) \rightarrow_{\mathbf{0}} B = \mathit{ce} \vdash \mathit{rprog} \dashv \Sigma_r' \rightsquigarrow \mathcal{C}\end{array}}{\Sigma_r \vdash \mathsf{fun}\ f(x : A) \rightarrow_{\mathbf{0}} B = \mathit{ce} \quad \mathit{rprog} \dashv \Sigma_r' \rightsquigarrow \mathcal{C}} \quad \text{RProg_purefun}$$

$$\frac{\begin{array}{c}\Sigma_c = \mathsf{Comp}(\Sigma_r) \quad \Sigma_c, f : A \rightarrow_{\mathbf{1}} B; \cdot, x{:}_{\mathbf{0}} A \vdash_{\mathbf{1}} \mathit{ce} \Leftarrow B \\ \Sigma_r, f : A \rightarrow_{\mathbf{1}} B \vdash \mathit{rprog} \dashv \Sigma_r' \rightsquigarrow \mathcal{C}\end{array}}{\Sigma_r \vdash \mathsf{fun}\ f(x : A) \rightarrow_{\mathbf{1}} B = \mathit{ce} \quad \mathit{rprog} \dashv \Sigma_r' \rightsquigarrow \mathcal{C}} \quad \text{RProg_impurefun}$$

$$\frac{\begin{array}{c}\Sigma_c = \mathsf{Comp}(\Sigma_r) \quad \Sigma_c, f : \tau \rightarrow_{\mathbf{S}} \tau'; \cdot, x{:}_{\mathbf{S}} \tau \vdash_{\mathbf{S}} \mathit{ce} \Leftarrow \tau' \\ \Sigma_r, \mathsf{fun}\ f(x : \tau) \rightarrow_{\mathbf{S}} \tau' = \mathit{ce} \vdash \mathit{rprog} \dashv \Sigma_r' \rightsquigarrow \mathcal{C}\end{array}}{\Sigma_r \vdash \mathsf{fun}\ f(x : \tau) \rightarrow_{\mathbf{S}} \tau' = \mathit{ce} \quad \mathit{rprog} \dashv \Sigma_r' \rightsquigarrow \mathcal{C}} \quad \text{RProg_specfun}$$

$$\frac{\begin{array}{c}\Sigma_r \vdash \mathit{Pf}_1 \multimap \mathit{Pf}_2\ [\mathbf{0}]\ \mathsf{wf} \\ \Sigma_r; \cdot \vdash q : \mathit{Pf}_1 \dashv \Delta \\ \Sigma_r; \Delta \vdash_{\mathbf{0}} \mathit{crt} \Leftarrow \mathit{Pf}_2 \dashv \Delta' \rightsquigarrow \mathcal{C} \\ \Sigma_r, f : \mathit{Pf}_1 \multimap \mathit{Pf}_2\ [\mathbf{0}] \vdash \mathit{rprog} \dashv \Sigma_r' \rightsquigarrow \mathcal{C}'\end{array}}{\Sigma_r \vdash \mathsf{fun}\ f(q) : \mathit{Pf}_1 \multimap \mathit{Pf}_2\ [\mathbf{0}] = \mathit{crt} \quad \mathit{rprog} \dashv \Sigma_r' \rightsquigarrow \mathcal{C} \wedge \mathcal{C}'} \quad \text{RProg_purerefined}$$

$$\frac{\begin{array}{c}\Sigma_r \vdash \mathit{Pf}_1 \multimap \mathit{Pf}_2\ [\mathbf{1}]\ \mathsf{wf} \\ \Sigma_r; \cdot \vdash q : \mathit{Pf}_1 \dashv \Delta \\ \Sigma_r, f : \mathit{Pf}_1 \multimap \mathit{Pf}_2\ [\mathbf{1}]; \Delta \vdash_{\mathbf{1}} \mathit{crt} \Leftarrow \mathit{Pf}_2 \dashv \Delta' \rightsquigarrow \mathcal{C} \\ \Sigma_r, f : \mathit{Pf}_1 \multimap \mathit{Pf}_2\ [\mathbf{1}] \vdash \mathit{rprog} \dashv \Sigma_r' \rightsquigarrow \mathcal{C}'\end{array}}{\Sigma_r \vdash \mathsf{fun}\ f(q) : \mathit{Pf}_1 \multimap \mathit{Pf}_2\ [\mathbf{1}] = \mathit{crt} \quad \mathit{rprog} \dashv \Sigma_r' \rightsquigarrow \mathcal{C} \wedge \mathcal{C}'} \quad \text{RProg_impurerefined}$$

$$\frac{\begin{array}{c}\Sigma_r \vdash \mathit{Pf}_1 \multimap \mathit{Pf}_2\ [\mathbf{S}]\ \mathsf{wf} \\ \Sigma_r; \cdot \vdash q : \mathit{Pf}_1 \dashv \Delta \\ \Sigma_r, f : \mathit{Pf}_1 \multimap \mathit{Pf}_2\ [\mathbf{S}]; \Delta \vdash_{\mathbf{S}} \mathit{crt} \Leftarrow \mathit{Pf}_2 \dashv \Delta' \rightsquigarrow \mathcal{C} \\ \Sigma_r, f : \mathit{Pf}_1 \multimap \mathit{Pf}_2\ [\mathbf{S}] \vdash \mathit{rprog} \dashv \Sigma_r' \rightsquigarrow \mathcal{C}'\end{array}}{\Sigma_r \vdash \mathsf{fun}\ f(q) : \mathit{Pf}_1 \multimap \mathit{Pf}_2\ [\mathbf{S}] = \mathit{crt} \quad \mathit{rprog} \dashv \Sigma_r' \rightsquigarrow \mathcal{C} \wedge \mathcal{C}'} \quad \text{RProg_specrefined}$$

$\boxed{\Sigma_r \vdash \sqcap \overline{\Delta} = \Delta}$

$$\frac{}{\Sigma_r \vdash \sqcap \Delta_0; \cdot = \Delta_0} \quad \text{NMerge_base}$$

$$\frac{\Sigma_r \vdash \Delta_0 \sqcap \Delta_1 = \Delta' \quad \Sigma_r \vdash \sqcap(\Delta'; \overline{\Delta}) = \Delta}{\Sigma_r \vdash \sqcap(\Delta_0; \Delta_1; \overline{\Delta}) = \Delta} \quad \text{NMerge_step}$$

```
Definition rules:        276 good    0 bad
Definition rule clauses: 597 good    0 bad
```